

The Riley Betts Erdős–Straus Programme: A Lean 4 Formalisation and Analytic-Geometric Investigation

Project Report on the Public Archive `erdos-strauss-main`

Revision **v0.2**

Prepared for Riley Betts Ltd

Compiled from the public git archive of the Erdős–Straus programme
(`erdos-strauss-main`, 21 August 2026)

22 August 2026

Abstract

This report documents the full contents of `erdos-strauss-main`, the public git archive of an extensive computer-assisted research programme directed at the Erdős–Straus conjecture. The programme does *not* claim a proof. Its concrete output is threefold. First, a substantial Lean 4 formalisation (thirteen source files, several thousand lines, axiom-audited) that reduces the conjecture to a single unproved analytic interface, proves a range of elementary and Brauer–Manin-theoretic facts about the associated affine surfaces, and certifies a finite verified range. Second, a sequence of analytic-number-theoretic partial results and honestly labelled negative findings on a covering-density reformulation of the conjecture, culminating in an unconditional exceptional-set bound stronger than Vaughan’s 1970 theorem, and a capstone diagnosis of why four independent completion attempts share a single obstruction. Third, a geometric investigation of the Bright–Loughran Brauer–Manin approach, which is shown to reduce, together with every other geometric mechanism considered, to the same analytic core. The archive also triages a sibling research effort and an external claimed proof, keeping only what survives scrutiny as library material. We describe the mathematical objectives, present the formal architecture in detail, discuss each analytic and geometric approach investigated together with its results (positive, partial, or negative), catalogue the accompanying numerical experiments, and give a complete inventory of every file in the archive together with the full bibliography the programme draws on.

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1 Introduction

1.1 The Erdős–Straus conjecture

The Erdős–Straus conjecture (ES), posed in 1948 and still open, asserts that for every integer $n \geq 2$ there exist positive integers x, y, z with

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Clearing denominators gives the equivalent Diophantine statement

$$\text{IsES}(n, x, y, z) : \iff x, y, z > 0 \text{ and } 4xyz = n(xy + yz + zx).$$

Classical work of Mordell, Schinzel, Aigner, Rosati, Yamamoto and others reduces the conjecture, via explicit polynomial identities, to the primes lying in a small number of residue classes modulo 840 that are quadratic-residue classes and hence not amenable to identity-based coverage: the six classes 1, 121, 169, 289, 361, 529 (mod 840), the smallest representative being $p = 1009$. The conjecture is exactly equivalent (Bloom’s covering-equivalence theorem, with ancestry in Nakayama, Rosati and Mordell) to an infinite covering-system statement: every n lies in a residue class $-a/c \pmod{4acd-1}$ for some positive integers a, c, d , or in one of finitely many auxiliary Type-II classes. Equivalently, writing $q = 4acd - 1$, solvability at n is the existence of a divisor $q \mid cn + a$ with $q \equiv -1 \pmod{4acd}$. The entire difficulty of the conjecture is concentrated in this covering statement for a density-zero

exceptional set: Vaughan proved in 1970 that the number of $n \leq x$ for which $4/n$ is *not* a sum of three unit fractions is $O(x \exp(-c(\log x)^{2/3}))$, and no exponent has been improved upon since.

1.2 What this archive is, and what it is not

The archive under report, `erdos-strauss-main`, is the public git repository of an AI-assisted mathematical research programme conducted under the auspices of Riley Betts Ltd, released under the MIT licence (with a single, separately attributed Apache 2.0 exception, Section 1.3). Its own front-door document, `erdos-strauss-programme.md`, states plainly:

*“This file is the programme brief for the git repository. It is not a theorem. **No proof of the Erdős–Straus conjecture is claimed or achieved.** The archive is kept so that the mathematical community can use the record.”*

This report inherits that discipline. Nothing below should be read as a claim that the Erdős–Straus conjecture has been proved, reduced to a solved sub-problem, or even reduced to a single named open conjecture that the community regards as tractable. What the archive contains is:

1. a genuine, if partial, **formal (Lean 4) development**, described in Section 3, that (i) is axiom-audited, (ii) proves the elementary reduction of ES to an explicit covering system and a single named analytic interface (`AnalyticSurvivorBound`), (iii) formalises a non-trivial slice of the Bright–Loughran Brauer–Manin theory for the associated affine cubic surfaces, and (iv) verifies the conjecture unconditionally for every $n < 100,000$ inside the kernel;
2. a sequence of **analytic partial theorems and precisely stated negative results** on a covering-density reformulation of the conjecture (Section 4), including an unconditional record improving on Vaughan’s 1970 exceptional-set bound, together with several careful “look and reject” investigations of recent analytic number theory (Zheng’s simultaneous-progressions theorem, Heath-Brown’s δ -method, Fouvry–Kowalski–Michel–Sawin monodromy independence, and a Maynard-type near-BV slab) against the specific obstruction the programme identifies (the *joint well-factorable CRT residue* at the Bombieri–Vinogradov wall), culminating in a capstone document that diagnoses the common failure mode of all four;
3. a **geometric investigation** (Section 5) of the Bright–Loughran Brauer–Manin obstruction theory for the Erdős–Straus surfaces, formalising several of their theorems, testing (and refuting) three candidate geometric proof mechanisms — Harpaz’s descent-fibration theorem, a real-compatible twist construction, and a triangle-boundary Campana/semi-integral existence principle — and arriving at the structural finding that every geometric mechanism considered reduces to the same analytic covering-density core as Route 1;
4. a **triage of a sibling research effort and an external claimed proof** (Section 6): a prior, separate covering-density assault is examined and mostly left out as a recorded negative, keeping only a ported Mathlib-style library theorem and a small Type-II identity dictionary; and a 2026 preprint claiming an elementary proof of ES is checked directly against the classical Mordell obstruction and found not to establish the covering claim it needs;
5. a substantial body of **numerical experimentation** in Python (Section 9), instrumenting every quantitative claim above with reproducible measurements at scales up to $x = 10^9$;
6. and a disciplined **research log** (the twenty-three sections §4a–§4w of `erdos-strauss-novel-structures-plan.md`, together with a chain of dated “postscript” entries in the programme brief) that records not only positive results but every investigated route that was tried and found wanting, with the specific obstruction named in each case.

A recurring stylistic feature of the archive, preserved throughout this report, is the sharp distinction between three categories of statement: results that are *proved* inside the Lean kernel (flagged [LEAN] below); results that are *claimed* on paper with a written mathematical argument but not machine-checked (flagged [CLAIMED]); and quantities that are *measured* computationally at finite, stated scales, which is evidence but not proof (flagged [MEASURED]). Distinct again are the several *negative* findings, where a specific published theorem or technique is checked against the programme’s target and found not to apply, for a stated structural reason (flagged [NEGATIVE RESULT]). This taxonomy is not a stylistic ornament: the archive’s central methodological claim is that most of the historical difficulty of the Erdős–Straus conjecture is explained by a single *k-budget invariant* (Section 4.1) that every one of these negative results instantiates in a different technical guise.

1.3 Authorship and provenance

The public archive is explicit about how the work was produced, and this report follows it in recording that explanation rather than paraphrasing it away. The programme brief states:

“Martyn Riley, of Riley Betts Ltd, facilitates the automation and the high-level strategy (novel routes, recorded negative results, documentation procedure for the agents). He makes no claim to expertise in number theory, combinatorics, or geometry. . . . The mathematical and formalization work is a collaboration of Anthropic agents, Grok 4.6, and Cursor agents, with Lean 4, Mathlib, and numerical test resources. It is not a record of human coauthorship of the mathematics.”

This report adopts the same framing throughout: where the text below refers to “the programme” producing a result, this denotes the automated research process described above, not a claim of individual human mathematical authorship. The repository is released under the MIT licence, copyright Riley Betts Ltd, with one stated exception: `Stormer.lean` (Section 3.13) is adapted from a public Mathlib pull request and remains Apache License 2.0, copyright Alexander Benjamin Worth Burns and Ravi Bajaj. A GitHub Actions workflow (`.github/workflows/ci.yml`) rebuilds the pinned Mathlib-dependent layer on every push, and a `CONTRIBUTING.md` records the archive’s pinning policy (Lean toolchain, Mathlib revision, lockfile, Python numerics) and reiterates the same scope restrictions as the mathematical documents themselves: no covering densification, no treatment of the external claims of Section 6 as resolved, no treatment of the library-only files as progress toward the load-bearing interface.

1.4 Objectives of the programme

Stated explicitly across the roadmap document `erdos-strauss-road-to-lean-qed.md` and the acceptance specification `erdos-strauss-conjecture-spec.md`, the programme’s objectives are:

1. to convert the folklore statement of ES into a machine-checked reduction whose only open content is a single, explicitly named analytic (or geometric) hypothesis;
2. to identify, as precisely as possible, *why* the conjecture has resisted proof for over 75 years — not merely to assert difficulty, but to locate a specific information-theoretic ceiling (the *k*-budget invariant) that every interval-intrinsic method is shown to respect, and to enumerate the input classes that are *not* capped by that ceiling;
3. to produce, along the way, genuine unconditional mathematics that stands on its own merit regardless of whether the summit is reached — in particular a new exceptional-set bound for the full covering problem, stronger than the 1970 record;

4. to formulate an **acceptance specification** (eight requirements, R1–R8, Section 8) that any candidate “load-bearing” conjecture C must satisfy to count as a genuine reduction of ES, and to test a wide slate of candidate conjectures against it (Section 7), several of which are refuted or frozen along the way;
5. and to record, as open mathematical questions addressed to the relevant published literatures rather than as correspondence with named individuals, a bounded set of questions in arithmetic geometry (integral points on log K3 surfaces, Section 5.10) and in analytic number theory (the half-dimensional/ vector-sieve literature, Section 4.10) that the programme’s own tools cannot resolve.

1.5 Structure of this report

Section 2 recalls the covering-system reformulation of ES that underlies the entire programme and the classical reductions it is built on. Section 3 gives a detailed, file-by-file account of the Lean 4 formalisation. Section 4 discusses the analytic covering-density route (“Route 1”), including the k -budget invariant, the sequence of theorems $T(1)/T(3)/T(A)/T(A)^+/E_lane/E_power$, the negative results obtained while attempting to complete the $T(3)$ lower bound, and their capstone diagnosis. Section 5 discusses the geometric Bright–Loughran route (“Route 2”), including the formalised Brauer–Manin theory, the Harpaz/Cao–Xu/twist/Campana investigations, the Schinzel separator programme, and the synthesis that the two routes are one route. Section 6 describes the triage of the prior Track-1 archive and the examination of an external claimed proof. Section 7 catalogues the eleven-plus named candidate conjectures the programme considered as the load-bearing statement C . Section 8 states the acceptance specification R1–R8 in full and gives the current scorecard. Section 9 describes the twenty Python scripts and their numerical findings in detail. Section 10 gives an overall assessment. Appendix A is a complete catalogue of every file in the archive. The full bibliography, drawn from `README.md` and cross-referenced throughout, is given at the end.

2 The covering-system reformulation

2.1 Witnesses and the covering criterion

The programme’s entire formal and analytic apparatus is built on a single elementary re-derivation, carried out independently in Lean (Section 3) and used throughout the analytic work. Suppose n admits parameters $a, c, d, m \in \mathbb{Z}_{>0}$ with

$$cn + a + m = 4acdm. \tag{*}$$

Call such a quadruple a *witness* for n , written $\text{Witness}(n, a, c, d, m)$. A short computation (the identity is proved as `witness_sound` in `ErdosStraus.lean`; Section 3.4) shows that a witness yields an explicit solution

$$4/n = \frac{1}{adm} + \frac{1}{n \cdot acd} + \frac{1}{n \cdot cdm}.$$

Writing $q = 4acd - 1$, equation (*) is equivalent to $q \mid cn + a$ with cofactor m ; when $c = 1$ this is the classical Type-I congruence $n \equiv -a \pmod{q}$. The *covering system* at level A consists of every cell (a, c, d) with $1 \leq a, c \leq A$, $1 \leq d \leq 5$; n is *covered* at level A if some cell in the box divides $cn + a$, and a *survivor* otherwise. Because a hit always yields a witness (this is exactly `ES.Covering.covering_sound`), the entire conjecture reduces to a statement about survivors:

$$\text{ES holds for } n \iff n \text{ is covered at some level } A.$$

The classical polynomial identities of Mordell, Aigner, Schinzel and others are simply particular, fixed, small-parameter witnesses (or one-parameter families of them); they dispose of every residue class modulo 840 except the six *hard classes* listed above. The programme’s Lean development formalises this reduction completely (Section 3.4.2), so that the open content of ES is exactly:

every sufficiently large prime in a hard class is covered by the level- $A(x)$ box, for a suitable growing schedule $A(x)$, and the finitely many small exceptions can be checked by computation.

This is the `HardLandingHypothesis` of the Lean development, and the single open interface the programme names is `AnalyticSurvivorBound` (Section 3.4.4).

2.2 The divisor-form duality

An equivalent and, for analytic purposes, more convenient coordinate system is the *divisor form*: n is covered by the cell (a, d) exactly when

$$q \mid n + 4a^2d, \quad q \equiv -1 \pmod{4ad},$$

for some divisor $q \geq 3$ of $n + 4a^2d$. The programme identifies this (`divisor_form_sound` in `ErdosStraus.lean`) as a re-coordination of the same covering box, with c playing the role of the free cofactor. In this form the covering condition is manifestly a statement about the prime (or divisor) factorisation of a shifted value $n + 4a^2d$, which is the natural language for sieve theory: it is exactly the setting of Iwaniec’s half-dimensional sieve and its relatives (Section 4). The quantity

$$\text{ClassRough}(p, a) : \iff p + 4a^2 \text{ has no divisor } q \geq 3 \text{ with } q \equiv -1 \pmod{4a}$$

(composite divisors included, not only aligned primes) is the Lean predicate that names one “slice” of the $d = 1$ covering box, and its joint density over a growing family of slices $a = 1, \dots, A$ is the object $S(A, x)$ studied throughout Section 4.

2.3 Classical background and the size of the hard classes

The six hard classes modulo 840 are exactly the classes that are quadratic residues modulo each of 8, 3, 5 and 7 simultaneously (equivalently, quadratic residues modulo 840), and it is a theorem of Mordell (via quadratic reciprocity) that no fixed polynomial witness family can cover a quadratic-residue class: this is why the classical reductions stop exactly at these six classes, and why the exceptional set of ES — if non-empty — is confined to them. The programme’s Lean development reproves this full reduction from scratch (Section 3.4.2); its analytic and geometric investigations are conducted entirely on this hard-class population.

3 The Lean 4 formalisation

3.1 Architecture: two orthogonal layer systems

The archive contains thirteen Lean source files together with a `lakefile.toml`, `lean-toolchain` and `lake-manifest.json` defining the build, plus a `.github/workflows/ci.yml` that rebuilds the pinned layer on every push. Eleven files sit at the repository root and are the active Lake library; two further files sit in an `archive/` subdirectory and are explicitly excluded from the build (Section 3.14). Two *different* stratifications named “Layer A/B(C)” occur in the documentation and must not be conflated (the archive itself flags this risk explicitly in `erdos-straus-novel-structures-plan.md`, §4s):

File-map layers (README.md). *Layer A* is the bare-Lean core — `ErdosStraus.lean`, `BrightLoughran.lean`, `ConicFiber.lean`, `TwistDescent.lean`, `SchinzelSep.lean`, `SchinzelDecide.lean`, `NoVieta.lean` — compiling under Lean $\geq 4.33.0$ with *no imports at all*, checkable with the bare `lean` binary and no `lake/Mathlib` dependency. *Layer B* is `ErdosStrausQR.lean`, `ErdosStrausBLRoute.lean`, and the two library-only additions of this revision, `Leochlon.lean` and `Stormer.lean` (Sections 3.12 and 3.13), all of which `import Mathlib`. The first two discharge the single deep mathematical input that Layer A leaves as an explicit interface (Hilbert/quadratic reciprocity); the latter two are unrelated library material kept for documentation reasons (Section 6) and carry no weight on the proof architecture.

Roadmap layers (erdos-strauss-road-to-lean-qed.md, §14). A different, orthogonal three-way split of the *proof architecture*: *Layer A* (formal algebra and the reduction to `HardLandingHypothesis`, fully compiled); *Layer B* (the still-open deep analytic or geometric theorem); *Layer C* (the finite certificate closing the range below the analytic threshold X_0 , whose lane is chosen among kernel `decide`, `native_decide`, or a future zero-knowledge-attested computation).

Both stratifications are in force simultaneously and are used side by side in the archive; this report follows the source in keeping them distinct, and uses “file-map Layer A/B” and “roadmap Layer A/B/C” where ambiguity could arise.

3.2 Build configuration and licensing

The Layer-B files are built with `lake` against a pinned revision of `mathlib4` (commit `1a0ef26d9624d64a1ad11853d17c30b8c32f2a10`) under `leanprover/lean4:v4.34.0-rc1` (`lean-toolchain`, explicitly documented as a release candidate rather than a moving nightly), declared in `lakefile.toml` (package `ErdosStraus`, eleven library roots). The Layer-A files require no such dependency and can be checked with a bare `lean` invocation per file, as documented in README.md. The lockfile `lake-manifest.json` is committed as part of the archive, and `CONTRIBUTING.md` directs that it be regenerated and committed alongside any toolchain or Mathlib bump; `.github/workflows/ci.yml` runs `lake exe cache get` and `lake build` on every push to guard against silent drift. Numeric scripts are pinned separately via `requirements.txt` (a single `numpy` version constraint). The repository is released under the MIT licence (`LICENSE`), copyright Riley Betts Ltd, with one exception: `Stormer.lean` (Section 3.13) carries its own Apache License 2.0 notice (`LICENSE-APACHE`), copyright Alexander Benjamin Worth Burns and Ravi Bajaj, as adapted from a public Mathlib pull request.

3.3 Axiom discipline

Every theorem in the file-map Layer-A core is accompanied, in README.md, by an explicit axiom audit obtained from Lean’s `#print axioms`. Three axiom profiles occur:

- **Axiom-free** (`decide`): theorems whose truth is certified by the Lean kernel’s own evaluator with no additional axiom, e.g. `primes_below_100_covered_kernel`, `lemma38_check_true`, `TwistDescent.descent`, `SchinzelSep.signed_9_5`, `SchinzelDecide.sep_9_5`, `SchinzelDecide.sep_13_7`, `SchinzelDecide.pos_4_5`.
- **Standard** (`propext`, `Quot.sound`, and sometimes `Classical.choice`): the great majority of theorems, including the main identity `witness_sound`, the conditional-QED assembly `conditional_qed_hard`, the conic-fibre correspondence `ConicFiber.fiber_to_divisor/divisor_to_fiber`, and `NoVieta.es_unique_x`.

- `native_decide` (adds `Lean.ofReduceBool`): used only for the large finite verifications that would be impractically slow under kernel reduction — `primes_below_10000_covered`, `hard_primes_below_10000_covered`, `okay_below_100000` (the verified frontier to $n < 100,000$), and the Brauer–Manin instance checks `thm12_instance_1009`, `witness_fingerprint_1009` in `BrightLoughran.lean`.

The file-map Layer-B files, including the two library additions of this revision, necessarily carry Mathlib’s own axiom footprint (in particular `Classical.choice` and `propext` pervasively, and `Quot.sound` where quotient types are used in Mathlib’s own development) and are not separately re-audited in `README.md`; the table in Appendix B reproduces the Layer-A audit exactly as published. The roadmap explicitly reserves a third, not-yet-implemented lane for closing the `native_decide` gap at larger scale: replacing its trusted-compilation axiom with a zero-knowledge-attested execution proof of the same checker (the lean-tee/SP1 pipeline referenced in `erdos-straus-novel-structures-plan.md`, Option 7/Track C), which is outside the scope of this archive but is named as the intended mechanism for scaling the verified frontier from 10^5 toward the classical computational record of 10^{17} (Salez, arXiv:1406.6307).

3.4 ErdosStraus.lean — the core reduction and covering system

This is the largest and most load-bearing file in the archive (3420 lines). It is organised into five successive strata, each building on the last.

3.4.1 Witness algebra and the finite verified frontier

- `IsES n x y z` is the cleared-denominator predicate defined in Section 2.
- `Witness n a c d m` is the covering condition (*).
- `[LEAN] witness_sound`: a witness for n yields the explicit ES solution $(x, y, z) = (adm, nacd, ncdm)$. This is the main algebraic identity the whole programme is built on, proved by a single omega-closed calculation after clearing denominators.
- `[LEAN] scale`: a solution for p lifts to any multiple of p — the standard observation that it suffices to prove ES for primes.
- `[LEAN] exists_prime_factor`: every $n \geq 2$ has a prime factor, proved self-containedly (no import of `Nat.exists_prime_and_dvd`), by strong induction with an explicitly defined `IsPrime` predicate and a proof of `IsPrime.dvd_or_dvd`.
- `[LEAN] LandingHypothesis`: every prime has a witness. `[LEAN] conditional_qed : LandingHypothesis → ErdosStraus` assembles the full conjecture from this one open hypothesis, via `witness_sound` and `scale`.
- `[MEASURED](native_decide) [LEAN] instances`: every prime, and separately every hard-class prime, below 10,000 has a witness with parameters ≤ 25 ; every prime below 100 is verified with the axiom-free kernel `decide`. A worked example, `es_1009 : IsES 1009 276 3027 92828`, is derived from the witness $(a, c, d, m) = (3, 1, 1, 92)$.
- An additional one-parameter family, the *one-slot* construction (`OneSlot`, Elsholtz–Tao Type I, complementary to the two-slot `Witness`), is proved sound (`oneSlot_sound`, `oneSlot_es`) and later folded into a *hybrid* covering predicate (`HybridCovered`) that is provably weaker to satisfy than the two-slot box alone.

3.4.2 The Mordell reduction to the six hard classes

A second stratum reproves, entirely inside Lean and by explicit one-line `omega` proofs from named polynomial witnesses, the classical elementary reduction:

- `[LEAN] es_of_not_one_mod_four`: ES holds outright for every $n \geq 2$ with $n \not\equiv 1 \pmod{4}$ — the class $n \equiv 3 \pmod{4}$ via witness $(1, 2, t+1, 1)$, and all even n by scaling the $n = 2$ solution.
- Seven further named polynomial witnesses discharge $n \equiv 2 \pmod{3}$; $n \equiv 5 \pmod{8}$; $n \equiv 3, 5, 6 \pmod{7}$; $n \equiv 7, 13 \pmod{15}$ (`w_two_mod_three`, `w_five_mod_eight`, `w_three_mod_seven`, etc.), each a single-line `omega` discharge of the witness equation.
- These are combined, via a `HardClass` predicate and a Presburger CRT lemma `mod840_hard` (again closed by `omega` alone), into `[LEAN] es_prime_not_hard`: *every prime outside the six hard classes satisfies ES unconditionally*.
- `[LEAN] HardLandingHypothesis` and `conditional_qed_hard` then restate the conditional QED so that the only remaining open hypothesis concerns primes $\equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ — exactly matching the classical state of the art, now machine-checked end to end.

3.4.3 The verified frontier below 100,000

`[LEAN] es_below_100000`, of type $2 \leq n \rightarrow n < 100000 \rightarrow \exists xyz, \text{IsES } n \ x \ y \ z$, is proved by strong induction combined with a self-certifying range check `okay`: at each n , either a proper factor is found (and the solution is obtained recursively by scaling), or a bounded covering witness is found directly. Because `okay` is self-certifying, no separate primality-completeness proof is required. This theorem uses the `native_decide` axiom lane described in Section 3.3 and is identified in the roadmap as the seed of the zero-knowledge-attested Track C: replacing `ofReduceBool` by a verified execution proof of the same checker, and scaling the bound from 10^5 towards 10^{17} , is described (but not implemented in this archive) as the intended lean-tee/SP1 pipeline.

3.4.4 The covering-system layer and the analytic interfaces

The fourth stratum (`namespace ES.Covering`, roadmap Steps 1–2) formalises the covering system of Section 2 as executable Booleans and proves everything about it that does *not* require analytic input:

- `coversB a c d n`, `coveredB A n` (a decidable Boolean scan of the level- A box), `Covered A n`, `Survivor A n` (with decidability instances).
- `[LEAN] covering_sound`: `Covered A n` implies the existence of a witness — the elementary half of the reduction, proved unconditionally for every level A .
- `[LEAN] survivor_typeI_aps`: a survivor avoids every Type-I arithmetic progression in the box — the “survivor bundle” used throughout the analytic discussion.
- `survivorCount A lo hi` and `[LEAN] no_survivors_of_card_eq_zero`: the *cardinality-to-QED bridge* — an explicit count of zero survivors in an interval kills every survivor there, by simple integrality.

- **The single load-bearing unproven target**, stated but deliberately left as a bare Prop (roadmap Step 2, “statement-first discipline”):

`AnalyticSurvivorBound (Alevel : Nat → Nat) (X0 : Nat) : Prop :=`

`∀ p, X0 ≤ p → IsPrime p → HardClass p → Covered (Alevel p) p.`

[LEAN] `hard_landing_of_interface` and `erdos_straus_of_interface` assemble this interface, together with a finite closure hypothesis below X_0 , into the full conjecture. *This definition is the precise formal target that Route 1 (Section 4) attempts, and fails within the scope of this archive, to discharge*; Section 4 explains exactly how far short of it the analytic results fall, and why.

- A parallel *divisor-form* interface, `DivisorLandingBound`, and the kernel equivalence `divisor_form_sound` (“the divisor criterion IS the covering box’s free- c slot”, `erdos-straus-novel-structures-plan.md` §4q), together with a family of bridge theorems (`dfi_dictionary_forward`, `dfi_dictionary`, `d1_landing`) connecting the two coordinate systems.

3.4.5 ClassRough, real characters, and the certificate architecture

The final and most intricate stratum of `ErdosStraus.lean` develops, entirely inside Lean, the *character-certificate* theory that underlies $T(A)$ and $T(3)$ (Section 4):

- `ClassRough p a`: $p + 4a^2$ has *no* divisor $q \geq 3$ (composite divisors included) with $q \equiv -1 \pmod{4a}$.
- A hand-rolled theory of *odd real characters* modulo 4, 8, 12 is built from scratch as completely multiplicative, periodic, ± 1 -valued functions: `chi4` (the character mod 4, $\chi_4(3) = -1$), `chi8`, `chi4chi8`, `chi3mod12`, `chi4chi3`, each proved multiplicative (`chi4_mul`, `chi8_mul`, `chi3mod12_mul`, ...) and periodic, and packaged as instances of a generic `OddRealChar` structure.
- [LEAN] `classRough_of_certifies` (the “certificate direction”): if a real odd character χ modulo $4a$ certifies $p + 4a^2$ — every prime factor of $p + 4a^2$ coprime to $4a$ lies in $\ker \chi$ — then `ClassRough(p, a)` holds. This is proved for general odd real characters via complete multiplicativity (`chi_eq_one_of_dvd`).
- The *converse* is proved as an exact biconditional at $a = 1, 2, 3$: [LEAN] `classRough_iff_chi4` ($a = 1$), `classRough_a2_iff_certificates` ($a = 2$, two characters $\chi_4, \chi_4\chi_8 \pmod{8}$), and `classRough_a3_iff_certificates` ($a = 3$, two characters $\chi_4, \chi_4\chi_3 \pmod{12}$), combined into the single **T(3) kernel theorem**

`classRough_123_iff_certificates`

stating that `ClassRough` jointly at $a = 1, 2, 3$ is *exactly* the $1 \times 2 \times 2$ union of the four resulting certificate assignments. This single theorem is the formal anchor of the entire $T(3)$ analytic argument in Section 4.3.

- A companion elementary stratum, `ThreeModFourFree` and the theorems `exists_three_mod_four_prime_factor`, `d1_a1_of_three_mod_four_factor`, `not_d1_a1_of_free`, identifies the $a = 1, d = 1$ escapee condition with “ $p + 4$ has no prime factor $\equiv 3 \pmod{4}$ ” — the classical sum-of-two-squares condition, and the base case that calibrates the whole certificate hierarchy.

3.4.6 The Henriot/Nair–Tenenbaum interface: an elementary $d = 1$ sieve, in Lean

The final ~ 800 lines of the file (from `coprime_four_of_odd` to `D1UniformNtBound` at the file’s end) carry out, entirely by elementary `omega`- and divisibility-level reasoning, a self-contained combinatorial sieve analysis of the $d = 1$, $a = 1$ lane — what the documentation calls “door (b)” of the k -budget investigation (Section 4.1). This is the only part of the archive’s analytic reasoning that is carried out *inside* the Lean kernel rather than on paper. Its architecture:

- An *inner/outer slot* dichotomy (`D1InnerSlot`, `D1OuterSlot`, `inner_or_outer`) partitions the aligned divisors of $p + 4$ by size relative to a threshold $2D$.
- A sequence of purely combinatorial “hit-spacing” lemmas (`no_three_hits_in_interval_len_q`, `at_most_two_hits_in_box`, `odd_hits_impossible_of_kq_gt_D`, `unique_large_aligned_landing`) bound how many aligned divisors of $p + 4$ can lie in a fixed-length window, by pure counting — no analytic input.
- A `SmoothTo` predicate and `outer_landing_has_factor_gt_two_D / outer_smooth_has_hub_prime` identify the two mutually exclusive escape mechanisms for an “outer” landing: either a genuinely large prime factor, or smoothness through a bounded “hub” prime.
- This is packaged into `D1NtExceptional`, `D1Remainder`, and finally a *named, unproved* target

`D1UniformNtBound (Dlevel : Nat → Nat) (X0 : Nat) (M : Nat → Nat) : Prop`

asserting that whenever a Nair–Tenenbaum/Henriot-type mean-value bound $M(p)$ on the number of class-rough slots stays strictly below the interval length $D_{\text{level}}(p)$, no $d = 1$ exceptional escape occurs. [LEAN] `D1Remainder_of_uniform_nt` and `erdos_straus_of_d1_uniform_nt` show, unconditionally, that discharging this one named hypothesis at the intended schedule $D \sim \exp(c\sqrt{\log x})$ (with M the Henriot implicit constant) would complete ES on the $d = 1$ lane. The hypothesis itself is explicitly *not* discharged in this archive; it is the Lean-level shadow of “door (b)” in the k -budget investigation (Section 4.1), and its status is identical to `AnalyticSurvivorBound`: named, wired, open.

This stratum is, in effect, a hand-formalised skeleton of a mean-value sieve argument, built to the point where only one external analytic input (a k -uniform Nair–Tenenbaum/Henriot bound) remains — precisely the “door (b)” investigation recorded on paper in `erdos-straus-novel-structures-plan.md` §4g–§4h (Section 4.1).

3.5 BrightLoughran.lean — the finite arithmetic heart of Theorem 1.2

This 407-line file formalises the elementary, finite-computation heart of Bright and Loughran’s Theorem 1.2 [13], isolating the single deep external input (Hilbert reciprocity over $\mathbb{Q}$) as an explicit `structure` interface.

- Serre’s 2-adic symbol data ([14]) is defined directly on residues: `eps r` ($= 1$ iff $r \equiv 3 \pmod{4}$) and `omg r` ($= 1$ iff $r \equiv \pm 3 \pmod{8}$).
- **The finite heart of BL Lemma 3.8.** In reduced 2-adic coordinates $(u_1, u_2, u_3) = (2^{s+e}r_1, 2^{s+e}r_2, 2^e r_3)$ with r_i odd and $s \geq 1$ (a shape forced by BL’s own valuation analysis), the 2-adic invariant of the Brauer class α is $(-1)^{f+g}$ with $f = \varepsilon(-r_1 r_3)\varepsilon(-r_2 r_3)$ and $g = s(\omega(r_1) + \omega(r_2))$. Because only residues mod 8 and $(2^s \bmod 8, s \bmod 2)$ matter, the general lemma reduces to a *finite* verification over $4^3 \times 4 = 256$ cases, executed by `lemma38Check` and proved true — *axiom-free* — by [LEAN] `lemma38_check_true : lemma38Check = true := by decide`.

- **InvariantData**: a **structure** packaging the four local invariants of α (at ∞ , at 2, at the odd prime $p \mid n$, and the product over all remaining good places) together with the single deep field

`reciprocity : invInf * inv2 * invP * invGood = 1`

— Hilbert reciprocity, left as an interface on bare Lean and discharged on a machine with Mathlib in `ErdosStrausQR.lean`.

- **[LEAN]** `bl_thm12_prime_anatomy`: given $\text{inv}_\infty = -1$ (the positive-octant invariant, BL Lemma 3.1), $\text{inv}_2 = 1$ (odd n , BL Lemma 3.8), and $\text{inv}_{\text{good}} = 1$ (BL Lemma 3.5), reciprocity alone forces $\text{inv}_p = -1$ — Yamamoto’s non-residue condition, i.e. exactly the covering-witness fingerprint of Section 2. `yamamoto_condition` restates this as a statement about the Legendre symbol directly.
- **[MEASURED]**(`native_decide`) **[LEAN]**: verified instances of Theorem 1.2 and its Yamamoto cross-check at the smallest hard-class prime, $p = 1009$: `thm12_instance_1009` and `witness_fingerprint_1009`, tying the geometric invariant directly to the covering witness `es_1009` proved in `ErdosStraus.lean`.
- BL Lemma 3.2 (the *unit-ratio lemma*: among three coordinates with distinct p -adic valuations, some ratio u_i/u_j is a p -adic unit) is proved in full, **[LEAN]** `lemma32`, together with the supporting Hilbert-symbol machinery (`hilbertOdd`, `hilbert2Odd`, `lemma34`, `lemma31/lemma31_pos` — the real-place invariant is -1 on the positive octant).

3.6 ErdosStrausQR.lean — the Mathlib discharge

A 170-line file importing `Mathlib.Tactic` and `Mathlib.NumberTheory.LegendreSymbol.QuadraticReciprocity`, this is the smallest Layer-B file and exists solely to connect the bare-Lean interfaces of `ErdosStraus.lean`/`BrightLoughran.lean` to genuine Mathlib number theory.

- **[LEAN]** `isPrime_iff_prime`: the hand-rolled `IsPrime` predicate coincides with `Nat.Prime`.
- **[LEAN]** `typeI_legendre / typeI_nonresidue_of_square_a`: the *Yamamoto fingerprint* — a Type-I witness with prime modulus $q = 4ad - 1$ forces $(p/q) = (-a/q)$ via the Legendre symbol; when a is itself a square and $q \equiv 3 \pmod{4}$, this forces $(p/q) = -1$, recovering the Schinzel/Mordell Type-I obstruction directly from Mathlib’s Legendre-symbol API.
- `quadratic_reciprocity_odd` instantiates Mathlib’s `legendreSym.quadratic_reciprocity` for the odd-place content of BL §4, and `thm12_instance_1009_mathlib`, `witness_1009_typeI_symbol`, `reciprocity_1009_11` reproduce the $p = 1009$ verified instances using genuine Mathlib reciprocity rather than the bounded-search `legendre` of Layer A.
- `lemma32_of_prime` specialises `BL.lemma32` to `Nat.Prime`.

Hilbert symbols themselves are not present in Mathlib, so `InvariantData.reciprocity` remains an interface even in this file; what is discharged here is the odd–odd quadratic reciprocity content it reduces to, which is the part Mathlib does supply.

3.7 ErdosStrausBLRoute.lean — the Bright–Loughran route, in full

At 3675 lines, this is the single largest file in the archive and the principal deliverable of the geometric investigation (Section 5). It develops, on top of the Layer-A/B interfaces above, the full affine $\mathbb{Z}$ -model of the Erdős–Straus and Schinzel surfaces, the Bright–Loughran octant invariant, and an extensive discharge of Hilbert reciprocity for the specific rational arguments the ES problem produces. Its contents, grouped by theme:

Character algebra and the hybrid covering bypass (plan §4p–§4q). `witness_legendre`, `typeI_d_slice`, `typeI_modulus_11_pairs`, `hard_prime_legendre_three`, and [LEAN] `survivor_legendre_three` formalise the finding that Type-I covering exclusions are unions of quadratic-residue *cosets*: a Legendre-symbol mismatch at a small modulus (e.g. $q = 11$) excludes an entire d -slice at once. `erdos_straus_of_character_covering` and `erdos_straus_of_hybrid_covering` package this as an alternative sufficient route to ES.

The affine $\mathbb{Z}$ -model (BL §2–§3). `IsSchinzelZ m n x y z` is the integer-point predicate for the affine cubic $m \cdot xyz = n(xy + yz + zx)$; `IsSchinzelZ 4 n` specialises to `IsESZ`. [LEAN] `isES_to_esZ/isES_of_esZ_pos`: natural-number solutions are exactly the positive-octant integer points, so [LEAN]

$$\text{erdos_strauss_iff_pos_octant} : \text{ErdosStrauss} \iff \forall n \geq 2, U_n(\mathbb{Z}) \cap C_+ \neq \emptyset.$$

[LEAN] `esZ_nonempty`: $U_n(\mathbb{Z})$ is nonempty for *every* $n \geq 2$ — even $n = 2t$ via the positive point $(2t, 2t, t)$, odd $n = 2k + 1$ via the classical mixed-sign point $\frac{1}{k} + \frac{1}{k+1} - \frac{1}{nk(k+1)}$, which lies *off* the positive octant (`esZ_of_odd_not_pos`). `schinzelZ_not_all_neg`: no all-negative point exists.

The Jahnel–Schindler/BL box (Lemma 3.10). [LEAN] `schinzelZ_min_le/esZ_min_le/es_pos_min_le`: every ordered integer point satisfies $m|u_{\min}| \leq 3n$, derived from the factor identity $(ry - s)(rz - s) = s^2$ with $r = mx - n$, $s = nx$ (`schinzelZ_factor`, `schinzelZ_coord_dvd`). The converse in the positive octant, [LEAN] `es_of_s2_divisor`, converts a divisor pair of s^2 at fixed x into a positive point; *existence* of such a pair as n varies is exactly the conjecture. A single explicit slice ($x = (n + 3)/4$, $r = 3$) is shown to occupy the octant for every $n \equiv 5 \pmod{8}$ (`es_five_mod_eight_of_repair`) and for those $n \equiv 1 \pmod{4}$ whose companion s has a prime factor $\equiv 2 \pmod{3}$ (`es_of_r3_slice`); it is shown, honestly, to *fail* at $n = 73$ (`r3_slice_obstructed_at_73`), and the file explicitly does not pursue chaining further moduli, describing that as “covering densification”.

The octant invariant and TUB-EP. `invInfES` is the real Hilbert symbol of α , cleared of the positive square u_3^2 . [LEAN] `invInfES_eq_neg_one_iff`: it equals -1 exactly on the same-sign octants (the all-negative octant being empty, this detects the positive component — BL Lemma 3.1 on the $\mathbb{Z}$ -model). [LEAN] `invInfES_of_esZ_mixed/invInfES_odd_mixed`: mixed-sign integer points have invariant $+1$ (BL Theorem 1.5 at the real place). The central named hypothesis of the geometric route is then

$$\text{TubEpHypothesis} : \forall n \geq 2, \exists u_1, u_2, u_3 > 0, \text{IsSchinzelZ } 4 \ n \ u_1 \ u_2 \ u_3,$$

with [LEAN] `erdos_straus_of_tub_ep` proved and [LEAN]

$$\text{tub_ep_iff_erdos_strauss}$$

establishing that `TubEpHypothesis` is *logically equivalent* to the full conjecture — a fact the report returns to repeatedly in Section 5, since a genuine reduction must have independent mathematical content beyond a bare restatement (requirement R2, Section 8). The $\mathbb{Z}$ -model facts any candidate existence principle may legitimately consume are packaged as a **structure** `TubEpConsumed` (nonemptiness, the box, the octant invariant, the mixed-sign-off- C_+ fact, and the positive-rational point $(n, n, n/2)$ that shows directly why Cao–Xu toric strong approximation cannot by itself imply ES).

Hilbert reciprocity, extensively discharged. The largest single portion of the file (roughly lines 1250–3000) proves Hilbert-symbol reciprocity, via Jacobi/ χ_4/χ_8 symbols, for every combination of sign and coprimality that the ES/Schinzel arithmetic actually produces: odd primes, odd coprime positives, negative odd coprimes, powers of 2, mixed signs, odd ratios, and specifically the BL 3.8 shape $(-2^s r_1/r_3, -2^s r_2/r_3)$ when the odd parts are pairwise coprime, together with a shared-odd-factor variant that removes the pairwise-coprimality restriction after cancelling common squares. Representative named theorems: `hilbert_reciprocity_odd_primes`, `hilbert_reciprocity_odd_coprime`, `hilbert_reciprocity_neg_odd_coprime`, `hilbert_reciprocity_two_pow`, `hilbert_reciprocity_neg_two_pow`, `hilbert_reciprocity_self`, `hilbert_reciprocity_neg_pos_odd_coprime`, `hilbert_reciprocity_odd_ratio`, `hilbert_reciprocity_es_ratio`, `hilbert_reciprocity_shared`, `hilbert_reciprocity_odd_integers`. These feed `[LEAN] thm12_discharged` (the bookkeeping $\text{inv}_\infty = -1$, $\text{inv}_2 = 1$, $\text{inv}_{\text{good}} = 1$, plus $\text{reciprocity} \Rightarrow \text{inv}_p = -1$) and `yamamoto_of_nat_solution`.

Lemma 3.8’s two-adic valuation analysis. A further self-contained block (lines ≈ 3030 –3600) proves the two-adic valuation bookkeeping BL Lemma 3.8 actually needs on *natural-number* solutions rather than the idealised reduced coordinates of Section 3.5: `es_v2_balance`, `exists_unique_min_pair`, `es_odd_two_adic_vals`, culminating in `[LEAN] lemma38_of_two_adic_shape` and `lemma38_of_nat_solution`, which derives the reduced (r_1, r_2, r_3, s, e) shape BL §3.3.4 requires directly from an arbitrary positive integer solution, closing the gap between Section 3.5’s finite lemma and genuine ES solutions. The file closes with `RepresentedX2p4Y2`, `odd_three_mod_four_free_is_form`, and `D1ShiftForm/d1_a1_iff_not_free`, tying the $a = 1$ certificate condition of Section 3.4.5 back to the classical sum-of-two-squares representation.

What this file does not claim. The file’s own header is explicit: it does not claim unconditional ErdosStraus or TubEpHypothesis; Theorem 1.8 is not formalised and is not treated as existence of a point; Cao–Xu is not treated as filling the octant; Harpaz is not applicable (Section 3.8); completeness of any covering system is not claimed.

3.8 ConicFiber.lean — the split-fibre theorem

This 194-line file is small, entirely self-contained, and proves the single geometric fact that determines the fate of every fibration-based attack considered in Section 5. Fixing the third ES coordinate t in $4txy = n(xy + tx + ty)$ and setting $A = 4t - n$, the surface equation is shown, by pure algebraic manipulation over $\mathbb{N}$ (no case split needed beyond positivity), to be equivalent to the identity

$$(Ax - nt)(Ay - nt) = (nt)^2,$$

proved as `[LEAN] fiber_identity`, built from the auxiliary lemmas `four_t_gt`, `fiber_relation` and `coord_exceeds`. Consequently *every* conic fibre of the Erdős–Straus surface is a *split* conic $uv = c$, for every $n \geq 2$ and every fibre t — not merely generically. Two further theorems make this constructive:

- `[LEAN] fiber_to_divisor`: a positive point on fibre t yields an explicit positive divisor pair (u, v) of $(nt)^2$ with $u \equiv v \equiv -nt \pmod{4t - n}$, and the coordinates recover linearly, $(4t - n)x = u + nt$.
- `[LEAN] divisor_to_fiber`: conversely, any such positive divisor pair in the correct residue class produces a positive point on fibre t .

Together these prove, unconditionally and for *all* $n \geq 2$, that along every conic fibration of the ES surface, the positive-point existence question *is* the divisor problem of Section 2 — the geometric and analytic routes are, at the level of this one theorem, the same object viewed in two coordinate systems. The immediate consequence, discussed at length in Section 5.3, is that Harpaz’s descent-fibration theorem for conic log K3 surfaces [17] cannot apply here: its hypotheses require *non-split* conic fibres carrying genuine Selmer structure, and none exists on this surface. The file’s axiom profile is the minimal `propext`, `Quot.sound` pair throughout.

3.9 TwistDescent.lean — the real-compatible twist and its Gate 3 failure

A 219-line file implementing the corrected form of candidate conjecture C5 (Section 7): the *real-compatible twist* $V_{n,d}$, obtained by adjoining $w^2 = d \cdot u_1 u_3$ ($d > 0$ squarefree) to the affine model U_n .

- `OnUn, IsTwist n d x y z w, PosRealLocus` set up the objects.
- `[LEAN] naive_cover_empty`: the *naive* α -trivialising cover $w^2 = -u_1 u_3$ has no real points at all over the positive octant (since $u_1, u_3 > 0 \Rightarrow u_1 u_3 > 0$) — the kernel form of the refutation of the original, naive version of candidate C5 (Section 7).
- **Gate 2** (the cheap half of requirement R1, Section 8): `[LEAN] descent`: an integral point of $V_{n,d}$ in the positive real locus projects to a positive-octant point of U_n . `[LEAN] lift`: conversely, every positive-octant point of U_n lifts to some twist (with $d = u_1 u_3$, not claimed squarefree-minimal).
- **Gate 3** (Section 5.5, the decisive negative finding): fibering $V_{n,d}$ by $u_2 = t$, an *explicit rational section* over $\mathbb{Q}(t)$ is exhibited and its cleared form proved as a Nat identity, `[LEAN] t_fiber_section_cleared`. A rational section means the generic fibre is *split*, so Harpaz’s Theorem 1.0.1 — whose engine requires non-split fibres — does not apply to this fibration either, for the same structural reason as Section 3.8.
- **Autopsy** (20 August 2026): the very same rational curve, reparametrised, is shown to also be a section of the projections $\pi = u_1$ and $\pi = w$ (`[LEAN] u1_fiber_section_cleared, u1_fiber_twist`) — changing the linear projection does not produce a genuinely non-split conic fibration of $V_{n,d}$.

This file is the kernel record of candidate C5’s refutation chain ($C5 \rightarrow C5_1 \rightarrow C5_2$, Section 5.5): a mathematically substantial idea, formally checked, and formally closed within the same investigation.

3.10 SchinzelSep.lean and SchinzelDecide.lean — the Schinzel separator suite

Two short files (84 and 268 lines) implementing the $S1/S1_1$ investigation (Section 5.7) into the generalised Schinzel family $m/n = 1/x + 1/y + 1/z$.

- `SchinzelSep.lean`: `OnSchinzel m n x y z` is the general cleared model. `[LEAN] signed_9_5`: the explicit mixed-sign integer point $(-5, 1, 1)$ satisfies $\frac{9}{5} = 1 + 1 - \frac{1}{5}$ (axiom-free `decide`). `[LEAN] no_pos_9_5`: for *ordered* $x \leq y \leq z$, no positive solution of $9/5$ exists — proved by a direct elementary bounding argument (the leading coefficient is forced to $x = 1$, then $y \in \{1, 2\}$, both residues fail), entirely by hand, no search.
- `SchinzelDecide.lean`: a genuinely *decidable* procedure `decideCplus m n`, searching $x \leq 3n$, $y \leq 2nx$ and recovering z from $nxy = zD$. `[LEAN] decideCplus_sound` and, more substantially, `[LEAN] decideCplus_complete` prove that this bounded search is both *sound* and *complete* for

positive-octant solvability of *any* fixed (m, n) — a genuine finite decision procedure, proved correct by an explicit chain of bounding lemmas (`exists_sorted`, and bounds `hb1–hb4` inside the completeness proof) rather than by brute enumeration alone. [LEAN] `sep_9_5`, `sep_13_7` (both axiom-free `decide`): the checker returns `false` on $9/5$ and on the companion separator $13/7$ (from the family $(2z - 1)/z$ at $z = 7$), yielding [LEAN] `no_pos_9_5` and `no_pos_13_7` as corollaries; `pos_4_5` is a positive control ($4/5 = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}$).

These two files jointly establish, unconditionally, that *mixed-sign integer occupancy of a Schinzel surface does not imply positive-octant occupancy* — the necessity check that drives the refinement of candidate conjecture C11 (Claim 5.1).

3.11 NoVieta.lean — no Markoff-type involution

A 156-line file proving the single structural fact that closes the “geometric mechanism inventory” of Section 5.9: the affine equation $mxyz = n(xy + yz + zx)$ is *linear* in each coordinate separately (unlike the Markoff equation $x^2 + y^2 + z^2 = 3xyz$, which is quadratic in each variable and hence admits the Vieta involution $x \mapsto 3yz - x$ underlying the Bourgain–Gamburd–Sarnak strong-approximation machinery [22]).

- `denomX m n y z = myz - ny - nz` is the fibre denominator for the x -coordinate; [LEAN] `linear_in_x`: $x \cdot \text{denomX} = nyz$ — the equation really is linear in x once (y, z) is fixed.
- [LEAN] `unique_x`: given a nonzero fibre denominator, x is uniquely determined by (y, z) — *there is no second root*, hence no Vieta-type involution and no Markoff-style correspondence orbit.
- [LEAN] `es_unique_x` specialises this to the ES equation ($m = 4$) on positive naturals.

This is a small file but a structurally important one: it is the formal reason, verified rather than merely asserted, that the entire Bourgain–Gamburd–Sarnak/Markoff-orbifold literature (Section 5.9) cannot be imported wholesale to the Erdős–Straus setting.

3.12 Leochlon.lean — a Type-II identity dictionary

A 95-line library file, new in this revision, importing `ErdosStraus.lean` together with Mathlib’s `Linarith` and `Ring` tactics. It formalises a single Egyptian-fraction identity reported for an external, informally documented construction (‘leochlon/erdstrau’; the archive notes that the originating URL was not reachable when first recorded, and treats the construction purely as a mathematical statement to be checked on its own terms):

$$(4b - 1)(4c - 1) = 4p\delta + 1, \quad \delta \mid bc.$$

- `LeochlonWitness p delta b c` (parameters p, δ, b, c) packages the hypotheses; [LEAN] `leochlon_key_identity` rewrites the defining equation over $\mathbb{Z}$ into the cancellation-free form $4bc = b + c + p\delta$.
- [LEAN] `isES_of_leochlon`: any such witness always yields the explicit Type-II Erdős–Straus triple $(pb, pc, bc/\delta)$ — i.e. p divides two of the three denominators, exactly the shape of the archive’s own `Witness` predicate (Section 3.4).
- [LEAN] `leochlon_of_q_succ_div_four` specialises to $b = (q + 1)/4$ when $q \equiv 3 \pmod{4}$, connecting the construction to the archive’s own small-modulus covering cells.

The file’s own header, and the companion note `erdos-straus-leochlon.md`, are explicit that this is a *dictionary* into `ES.IsES` and nothing more: the identity is not automatically a covering-landing witness at any particular (a, c, d, m) , and it is not treated as a gateway or finite-catalogue covering construction. Section 6 discusses the provenance and scope of this file in full.

3.13 Stormer.lean — a ported library theorem, out of scope for the proof

An 855-line file, new in this revision and licensed separately from the rest of the archive (Section 1.3), proving Størmer’s 1897 theorem [78] on consecutive smooth numbers, adapted from a public Mathlib pull request (#42040) to build outside Mathlib’s own `module` root. The file’s own header states plainly that it is “library only” and “not on the T(A)/geometry QED path”. Its mathematical content: given a positive solution of the Pell equation $x^2 = 1 + dy^2$ whose y -coordinate has no prime factor outside a fixed finite set together with the Pell parameter d , the solution is forced to be the *fundamental* one ([LEAN] `eq_fundamental_of_primeFactors_y_subset_parameter` and `eq_of_primeFactors_y_subset_parameter`, built on an explicit prime-divisibility lemma for the Lucas-sequence coordinate `pellY`, [LEAN] `exists_prime_dvd_pellY_not_dvd_parameter`). This is then packaged into a three-valued “exponent code” on s -smooth numbers (`exponentCode`, `consecutiveCode`) and used to bound the number of consecutive pairs of s -factored naturals: [LEAN] `card_consecutive_factoredNumbers_le_sub` gives an explicit bound of $3^{|s|}$ minus a correction, and [LEAN] `finite_consecutiveFactoredNumbers` concludes finiteness of the set of such consecutive pairs for any fixed finite prime set s . None of this material is used anywhere else in the archive; it is included, and separately licensed, purely as a documented piece of provenance from the sibling investigation discussed in Section 6.

3.14 Frozen and legacy files

Two further Lean files are present but explicitly marked as non-canonical by `README.md`, and are kept in an `archive/` subdirectory excluded from the `lakefile.toml` build: `archive/ErdosStraus-core.lean` (2073 lines) is described as “a frozen snapshot of an earlier covering file, not a Lake root”, and `archive/ErdosStraus-mathlib-project.lean` (2480 lines) as “a duplicate of `ErdosStraus.lean`”. Both share the same header and opening definitions as the canonical `ErdosStraus.lean` (Section 3.4) but represent earlier development states; a short `archive/README.md` states explicitly that they are not Lake roots and should not be imported. This report does not describe them further.

3.15 Summary of what is, and is not, proved

Collecting the strata above, the unconditional content of the Lean formalisation is:

- the full elementary reduction of ES to `HardLandingHypothesis` (six hard classes mod 840), with every classical polynomial-witness case machine-checked;
- soundness of the covering system at every level (`covering_sound`) and the cardinality-to-QED bridge;
- unconditional verification of ES for every $n < 100,000$;
- the finite arithmetic heart of Bright–Loughran’s Theorem 1.2 (Lemma 3.8’s 256-case table), verified instances at $p = 1009$, and an extensive discharge of Hilbert reciprocity for the rational arguments the ES problem actually produces;
- the $\mathbb{Z}$ -model nonemptiness, box bound, and octant-invariant theorems, and the equivalence `TubEpHypothesis` $\iff$ `ErdosStraus`;
- the split-fibre theorem and its exact divisor correspondence, for every $n \geq 2$ and every conic fibre;

- the Gate 2/Gate 3 twist-descent theorems, including the rational-section proof that closes candidate C5;
- the Schinzel-separator decision procedure and its verified instances at 9/5 and 13/7;
- the no-Vieta uniqueness theorem;
- and, as library material carrying no weight on the reduction, a checked Type-II identity dictionary (`Leochlon.lean`) and a ported Mathlib-style theorem on consecutive smooth numbers (`Stormer.lean`), both discussed further in Section 6.

What remains explicitly, and by design, unproved and named as such are exactly two interfaces: `AnalyticSurvivorBound` (equivalently `DivisorLandingBound`, equivalently `D1UniformNtBound` on the restricted $d = 1$ lane) on the analytic side, and `TubEpHypothesis` on the geometric side — with the kernel theorem `tub_ep_iff_erdos_straus` establishing that the latter has, at present, no content beyond a restatement of the conjecture itself. Neither interface is discharged anywhere in this archive.

4 Route 1: the analytic covering-density approach

4.1 The research log and the identification of the k-budget invariant

Sections §1–§4l of `erdos-straus-novel-structures-plan.md` record a single day (18 August 2026) of rapid, iterated attempts on a *covering density lemma* — an upper bound on the density of integers avoiding every congruence class in a growing covering box — together with the honest documentation of where each attempt stalled. The log is unusual for a research document in that it preserves refuted intermediate claims rather than silently correcting them; this report follows that discipline and reconstructs the chain in order, since the final negative result (the k-budget invariant) is only meaningful in light of what it subsumes.

The ladder of reformulations (§4a–§4d). An initial *Landing Lemma* (roots of the quadratic congruence $\nu^2 \equiv -pd \pmod{q}$ landing in a divisor-constrained position) is identified as equivalent, via the classical dictionary of quadratic forms and ring class fields, to a Duke–Friedlander–Iwaniec-style equidistribution statement ([69]; this is “Option 9/Option 5 fusion” in the plan’s own numbering). A first quantitative version, the *Covering Density Lemma*, conjectures $u(A) \ll A^{-2-\varepsilon}$ for the density of survivors of the level- A box, calibrated against a measured constant $\kappa \approx 0.139$ (the exponent of $\log^2 A$ in the covering mass). A *partial theorem* is then proved rigorously by restricting to prime cofactor m and using exact CRT independence across distinct primes plus Bombieri–Vinogradov, giving an unconditional power-law density bound with exponent $\kappa^* \approx 0.29$ — the first genuine theorem of the investigation, and a template for everything that follows.

The Suen conditioning scheme (§4e). Conditioning on all primes $\leq T$ turns the survivor-density problem into a product space over large-prime coordinates, to which *Suen’s correlation inequality* [29, 30] applies directly:

$$u \leq \exp(-\mu + \Delta e^{2\delta}),$$

with μ the total conditioned mass, Δ the mass on dependent pairs, and δ the maximum neighbour mass. Measured gates at toy scale ($A = 24\text{--}48$) show that a small-prime “hub” (the primes 3, 7, 11, 13) is the entire source of dependence, and that conditioning it away reduces Δ, δ to

$O(\mu^2/T), O(\mu/T)$ — an elementary second-moment bound, later made fully rigorous as Lemma SM (Section 4.7). This is identified as, in principle, a route to the *full* super-polynomial covering-density bound $u(A) \leq \exp(-(1 - o(1))\kappa \log^2 A)$.

The Vaughan barrier and its correction (§4f–§4h). A first (later corrected) attempt to transfer this density bound into an exceptional-*count* bound over a growing interval $[x, 2x]$ derives a sieve-dimension constraint $\theta < 1/3$ for a covering level $A = \exp((\log x)^\theta)$, and — remarkably — recovers *exactly* Vaughan’s 1970 exponent $\exp(-c(\log x)^{2/3})$ at the boundary $\theta = 1/3$, independently from three unrelated technologies (the fundamental lemma, the large sieve, and Vaughan’s original exponential-sum method). This “triple convergence” is initially read as strong evidence for a genuine barrier. A subsequent, more careful trace of the relevant Nair– Tenenbaum/Henriot machinery (§4g–§4h, “door (b)”; this is the analysis formalised in Section 3.4.6) discovers that the original barrier arithmetic *conflated total sifted mass with sieve dimension*. Corrected, the achievable cap is not $\exp(-c(\log x)^{2/3})$ but a genuine *power saving* $x^{-c\kappa}$, i.e. an unconditional exceptional-set bound $O(x^{1-c})$ **strictly stronger than Vaughan**. This correction — explicitly flagged as such in the document, with the original §4f retained unedited as a record — is the origin of the E₋power theorem (Section 4.7).

The k-budget invariant, in final form (§4k–§4l). The corrected barrier is then re-derived a *third* independent way, via a circuit-complexity/“fooling” argument (intervals fooling a depth-2 CRT formula, in the style of Braverman–Bazzi-type constructions), and the three derivations are triangulated into a single invariant:

***The k-budget invariant.** The interval-intrinsic independence budget is $k \cdot \log A \leq \log x$, while QED-precision requires effective independence $k \gtrsim \text{mass} > \log x$. Every method consuming only interval-intrinsic randomness — sieves, the large sieve, Shiu-type transfers, k-wise-independence fooling — is capped by this inequality.*

The invariant is not claimed as a theorem (the plan calls it “conjectural, proof-shaped”), but it organises the entire remainder of the programme: it identifies exactly two input classes that are *not* capped by it — genuinely spectral/bilinear arithmetic input (Kloosterman/DFI-type cancellation), and algebraic rigidity used non-statistically (the reciprocity/character-coset structure of Section 4.2) — and every subsequent “look” investigation in Section 4.4 is, explicitly, a test of whether some named modern technique supplies input from one of these two classes at the specific modulus where the programme’s own argument stalls.

The resulting statement of the residual open problem is named **H₋ES** (Survivor Level of Distribution): at level $A = \exp(c\sqrt{\log x})$ with $\kappa c^2 > 1$, the survivor count in $[x, 2x]$ should satisfy $\#S \ll x e^{-(1-o(1))\kappa \log^2 A}$. This is, precisely, **AnalyticSurvivorBound** restated analytically; it is not proved anywhere in the archive, and the k-budget invariant is the programme’s own explanation of why it should be hard.

4.2 T(A): the joint density of class-roughness at fixed width

The central working object of the programme’s second phase (20–21 August 2026, **erdos-strauss-T-A.md**) is a reformulation of the density question in terms of the **ClassRough** predicate of Section 3.4.5. For p ranging over hard primes, write

$$\rho_a(x) = \Pr[\text{ClassRough}(p, a)], \quad S(A, x) = \Pr[\text{ClassRough}(p, a) \text{ for all } 1 \leq a \leq A],$$

and

$$\widehat{C}(A, x) = \frac{S(A, x)}{\prod_{a=1}^A \rho_a(x)}$$

for the *selection constant*. The conjectural statement $T(A)$ is that, for each fixed A , $S(A, x) \asymp C(A) \prod_a \rho_a(x)$ with $C(A)$ an explicit, trackable sequence, and the programme’s stated exposure is

$$\limsup_{A \rightarrow \infty} \frac{\log C(A)}{\log^2 A} < \kappa = 0.139$$

(the same constant as Section 4.1), *together with* the requirement that extra covering width continue to contribute genuine mass rather than becoming what the archive calls *dummy covering* — the two “die conditions” tracked throughout.

The certificate architecture. `ClassRough( $p, a$ )` is shown (Section 3.4.5) to be *certified* by any real odd character $\chi \bmod 4a$ all of whose non-trivial values on prime factors of $p + 4a^2$ avoid the forbidden class — this is the exact, kernel-checked direction. The converse (every class-rough p has *some* certifying character) is proved as an exact biconditional only at $a \in \{1, 2, 3\}$ (and, by a parallel elementary argument, at $a = 6$) — precisely the values of a for which the unit group $(\mathbb{Z}/4a\mathbb{Z})^\times$ is an elementary abelian 2-group. [MEASURED] At general a the converse is an ω -gap, not a near-equivalence: at $x = 10^6$, $A \leq 40$, the certificate covers only 84.6% of class-rough survivors, with the shortfall growing sharply with the number of coprime prime factors $\omega(N)$ of the shifted value (100% at $\omega = 1$, falling to 35% at $\omega = 4$); this is the *defect layer*, shown (`c4_defect_layer.py`) to be exactly the combinatorial phenomenon that -1 can lie in the subgroup generated by the prime residues of N without being a literal subset product, which is possible precisely when $\lambda(4a) \nmid 2$, i.e. for $a \notin \{1, 2, 3, 6\}$.

4.3 $T(3)$: a two-sided bracket, upper half claimed

The **$T(3)$ kernel theorem** (`classRough_123_iff_certificates`) makes $a = 1, 2, 3$ exact: `ClassRough` jointly on these three slices is precisely the union of four certificate assignments built from the characters $\chi_4 \pmod{4}$, $\chi_4, \chi_4\chi_8 \pmod{8}$, and $\chi_4, \chi_4\chi_3 \pmod{12}$. [MEASURED] At $x = 10^6$ this is verified against 2,370 hard primes with an exact match (656 class-rough on all three slices, 656 in the certificate union).

Claim 4.1 ($T(3)^+$, [CLAIMED]).

$$S(3, x) \ll (\log x)^{-3/2} \quad (x \rightarrow \infty),$$

with an effective, explicit implied constant.

The proof (given in full in `erdos-strauss-T-3.md`) is a genuine, if not machine-checked, Selberg upper-bound sieve of dimension $3/2$ on the six hard classes, sieved to level $z = x^{1/4}(\log x)^{-B}$, with Bombieri–Vinogradov supplying the remainder term at the required level $z^2 \leq x^{1/2-}$. The dimension $3/2$ arises because each of the three certificate kernels excludes a density- $1/2$ set of primes q , and the three densities sum (up to $O(1)$ collision corrections, verified to be exactly zero for the relevant small moduli) to $3/2$.

Claim 4.2 ($T(1)$, [CLAIMED], by citation). `ClassRough` at $a = 1$ is exactly the χ_4 -certificate, i.e. $p = x^2 + y^2 - 4$ for hard $p \equiv 1 \pmod{4}$, and by Iwaniec [47] together with the congruence-layer extension of Fuchs–Hsu–Rickards–Schindler–Stange [67], $S(1, x) \asymp (\log x)^{-1/2}$.

This one-slice asymptotic calibrates the exponent $\kappa = 1/2$ per certificate that the dimension-count above extrapolates.

The matching lower bound: a three-step plan with a named stall. The programme is explicit that the two-sided $T(3)$ bracket is *not* claimed. The plan for $c_- > 0$ is: **(a)** Iwaniec’s half-dimensional sieve [48] gives a genuine lower bound per certificate slice (dimension $1/2 < 1$ evades the parity/sifting-limit problem); **(b)** the Brüdern–Fouvry vector-sieve inequality [49, 50] (with Nath–Xie [66] as the two-component template) combines lower-bound weights across the three simultaneous kernels; **(c)** *completion* — passing from “no bad factor below z ” to the genuine certificate condition requires controlling factors $q > x^{1/2}$, which sits beyond Bombieri–Vinogradov and is where the argument can, and the programme believes does, stall. Two *escapes* are recorded for the eventual paper’s outlook rather than claimed: a weaker c_- (a positive proportion of the expected order, via “almost-certificates”), or switching to the divisor-sum detector $r_\chi(n) = \sum_{d|n} \chi(d)$ and a dispersion-method (Linnik [52]) attack. The sharp remaining mathematical question, stated precisely and repeatedly across `erdos-strauss-T-3.md`, `erdos-strauss-sieve-desk.md` and `erdos-strauss-novel-structures-plan.md` §4w, is:

Does a joint well-factorable weight framework exist for a CRT residue $\alpha(d_1, d_2, d_3) \bmod \text{lcm}(d_1, d_2, d_3)$ that depends on the summed triple, and if not, can one be built for this specific three-kernel case (conductors 4, 8, 12)?

4.4 Four negative results at the completion stall

Between 21 August 2026 and the archive’s final state, four separate “look and reject” investigations were carried out against exactly this stall, each testing a specific recent piece of analytic machinery. All four are honestly negative, and each pins down precisely why.

(i) Zheng’s simultaneous arithmetic progressions [NEGATIVE RESULT] (`erdos-strauss-T-2.md`, `zheng-wellfactorable.md`). Zongkun Zheng’s theorem on primes in two simultaneous arithmetic progressions [54] is tested as a two-modulus ($T(2)$) mean-value input. A first technical obstacle (`zheng-wellfactorable.md`) is that the natural weight $\lambda_d = \chi(d)\mathbf{1}_{d \leq D}$ is *not* automatically well-factorable in Zheng’s sense (Definition 1.1 is a Dirichlet-convolution statement, not a pointwise factorisation), a subtlety resolved by isolating the squarefree part. Substantively: after switching, both divisors d_1, d_2 lie in $[0, \sqrt{x}]$, and Zheng’s Theorem 1.1 restricts one leg to $\theta \leq 7/36$. [MEASURED] On a 181×181 grid of the switched exponent plane, the symmetric cell $d_1 \approx d_2 \approx \sqrt{x}$ — exactly the cell the completion step needs — lies in *neither* Bombieri–Vinogradov nor Zheng’s Theorem 1.1 or 1.2; the union of ordinary BV and Zheng covers only 0.518 of the relevant plane, and an “uneven” reroute that leaves one of the three $T(3)$ kernels in a Type I/polylog slot and applies Zheng to the remaining pair still fails on the cell $(0, 1/2, 1/2)$ (`t2_zheng_ranges.py`). **Verdict:** range no-go; not $T(2)^+$; not $T(3)$ progress.

(ii) Heath-Brown’s δ -symbol method [NEGATIVE RESULT] (`erdos-strauss-T-3-delta.md`). Heath-Brown’s dissection identity [68] is tested as a way to detect the moving CRT congruence jointly rather than sequentially, hoping the Farey/dissection parameter Q_δ supplies a genuinely extra degree of freedom. A short lemma (“Farey- k incompatibility”) shows that at the Bombieri–Vinogradov wall ($M + N = 1$, $Q_D = 1/2$), opening a bilinear cell against the dissection modulus forces $Q_\delta > 1/2$, while completing the residual AP-index sum forces $Q_\delta \leq 1/2$ — the same identity as the Phase 1 stall, merely repackaged (`t3_delta_ranges.py`). At the specific uneven cell $(0, 1/2, 1/2)$ this particular incompatibility is vacuous (the AP index is absent there), but no positive saving is obtained up to $D \sim x$ or $D \sim x^{3/2}$ either. **Verdict:** does not avoid the wall; no range gained; not $T(3)$ progress.

(iii) **Joint monodromy independence** [**NEGATIVE RESULT**] (`erdos-strauss-T-3-monodromy.md`). The 2026 Fouvry–Kowalski–Michel–Sawin theorem on bilinear forms with trace functions [73] is tested as a route to square-root cancellation in the product $\chi_2(d_2)\chi_3(d_3)K(\cdot; D)$ for two moving quadratic-character moduli. Four structural mismatches are identified: the relevant kernel is not a trace function of a single ℓ -adic sheaf on a product variety over a *fixed* finite field (the modulus D moves with the summation point); the individual characters χ_2, χ_3 are rank-1 Kummer sheaves with abelian monodromy, not the required “gallant” (irreducible, quasisimple) type; after switching, a *prime* modulus in this family cannot exceed $\sqrt{x}$, landing exactly on the BV wall, where the relevant length inequalities fail; and a *composite* modulus (two prime factors) has monodromy of type $\mathrm{SL}_2 \times \mathrm{SL}_2 \rightarrow \mathrm{SO}_4$, which FKMS explicitly classify as *not* gallant. **Verdict:** the theorem’s hypotheses are not met on any reachable cell; no range past $x^{1/2}$ is obtained; not T(3) progress.

(iv) **The r_χ -to-Kloosterman bilinear reduction** [**NEGATIVE RESULT**] (`erdos-strauss-T-3.md`, **Phases 0–4**). Escape 2 of the three-step plan (Section 4.3) is carried through explicitly: the weighted count $W(x) = \sum_p r_{\chi_1}(p+4)r_{\chi_2}(p+16)r_{\chi_3}(p+36)$ is expanded, the inner congruence system is solved via a moving CRT residue $\alpha(d_1, d_2, d_3)$, and — after opening Type II via additive characters — the remainder is reduced to an explicit trilinear Kloosterman sum, of exactly the shape treated by three 2025–26 papers: Pascadi [63], Milićević–Qin–Wu [64], and Blomer–Pascadi [65]. A short lemma (“no bilinear cell at the wall”) shows that at the Bombieri–Vinogradov boundary itself $\delta = 0$ (writing the modulus as $x^{1/2+\delta}$) there is provably no bilinear cell at all — one Type II factor necessarily folds to a complete residue system. [MEASURED] Testing the saving ranges of all three cited theorems against the dual-interval lengths available just past the wall: Pascadi’s Corollary 1.4 length constraint is satisfied but its *saving* range is not reached until $\delta \gtrsim 1/3$; Blomer–Pascadi’s genuine saving over the trivial Weil bound with balanced dual lengths first appears only at $\delta \approx 0.367$ ($Q \approx x^{0.87}$); MQW’s theorem uses the wrong (product-argument, not AP-Poisson) kernel shape and its length hypotheses require $Q \geq x^{4/5}$ (`t3_kloosterman_ranges.py`). **Verdict:** no-go at the stall; the entire admissible-saving gap $x^{1/2} < Q < x^{0.87}$ remains untouched by any cited 2025–26 result; not a third escape.

Remark 4.3. The four negative results above are methodologically of a piece: each identifies a specific, named, currently-published theorem that a naive reading might expect to bridge the Bombieri–Vinogradov wall for this problem, and each is shown, by an explicit range or hypothesis check rather than a vague appeal to difficulty, not to reach the cell the completion step of T(3) needs. None of the four claims to have proved a barrier theorem; all four are consistent with — and are read by the programme as further evidence for — the k -budget invariant of Section 4.1, since all four available techniques supply either interval-intrinsic input (capped) or genuinely spectral input that happens not to reach far enough in this specific family at this specific modulus. Section 4.5 draws out the common structural reason.

4.5 The varying-modulus correlation gap: a capstone diagnosis

A later document, `erdos-strauss-varying-modulus-gap.md`, steps back from the four individual negative results of Section 4.4 and asks what they have in common. It adds a fifth, informally recorded route to the comparison — an attempt using Maynard-type uniform-residue sieve technology (Maynard I–III) to reach a thin near-Bombieri–Vinogradov slab just past the wall, which the document notes was never filed as a separate look because it fails for the same reason as the Zheng reroute: any window inside that thin slab still forces sacrificing one of the three kernels, which reduces to the already-closed two-modulus cell of Section 4.4(i). With this fifth route included, the document states its finding as an absence rather than a construction: after Phase 1 of `erdos-`

strauss-T-3.md (Section 4.3), what remains is square-root cancellation, uniform in d_2, d_3 , of

$$\sum_{d_2, d_3} \chi_2(d_2) \chi_3(d_3) K(\cdot; D), \quad D = \text{lcm}(d_2, d_3) \asymp d_2 d_3,$$

where both d_2 and d_3 run up to (and, before switching, past) $\sqrt{x}$, so that D itself is the *jointly growing, multi-factor* object — not a fixed quantity relative to which something else varies.

Claim 4.4 (The two-shape classification, [CLAIMED]). *Every method surveyed by the programme, across all five routes, falls into exactly one of two shapes: (1) **fixed residue, growing modulus** — Bombieri–Vinogradov, BFI well-factorable estimates, and Maynard I–III, which all want a residue a held independent of q as q grows, whereas the $T(3)$ residue $\alpha(d_2, d_3)$ moves with the modulus by construction; or (2) **fixed modulus (field), varying argument** — the δ -symbol’s Farey dissection past the BV wall, and the entire ℓ -adic trace-function lineage (Katz [72]; Fouvry–Kowalski–Michel on sums of products [75] and on trace functions over the primes [74]; Kowalski–Michel–Sawin [60]; Fouvry–Kowalski–Michel–Sawin [73]), each of which fixes a base field $\mathbf{F}_q$ (or a prime modulus q) and lets an auxiliary argument, tuple, or family index vary instead.*

No route surveyed supplies cancellation when the modulus itself is the jointly growing, multi-factor object — this is exactly the FKMS failure mode already identified in Section 4.4(iii), and the document traces the same cut through the wider family tree behind it, including Ricotta–Royer’s Kloosterman paths for prime-power moduli p^n with $n \geq 2$ fixed (only $p \rightarrow \infty$ varies) [76], and a partial, harder adjacent case (Kowalski, uniform monodromy as both the field and a conductor grow, for elliptic-curve twists [77]) that the document reads as structurally closer to what $T(3)$ would need but not the same object. The open question is posed precisely:

Does there exist — or can one construct — a Goursat–Kolchin–Ribet-type independence statement for a family of Kloosterman-type (or gallant) sheaves in which the modulus is the outer, growing, multi-factor summation variable, rather than the fixed base field over which a single sheaf’s argument ranges?

The document explicitly does not claim the gap is permanent — only that the specific family tree already engaged by the programme does not contain a citation that closes it — and it notes one untested avenue as a plausible testbed rather than a claimed result: the function-field analogue $\mathbf{F}_q[t]$, where Weil-quality bounds are unconditional, so that independence across several jointly growing polynomial moduli might be stateable and checkable before any attempt over $\mathbb{Z}$. This document, more than any single one of the four negative results it synthesises, is the programme’s clearest statement of exactly where the $T(3)$ lower bound’s difficulty is currently understood to live.

4.6 $T(A)^+$: a uniform Selberg bound, and E_- lane

Independently of the $T(3)$ completion question, the programme proves a genuinely unconditional upper bound at *every* fixed covering width A , generalising the $T(3)^+$ argument.

Claim 4.5 ($T(A)^+$, [CLAIMED]). *For each fixed $A \geq 1$, writing $\beta(A) = \sum_{a \leq A} 1/\varphi(4a)$ for the joint sieve dimension (the number of distinct forbidden residues, which residue collisions can only decrease — verified computationally to be exactly zero for $A \leq 200$),*

$$S(A, x) \ll C_{\text{sieve}}(A) (\log x)^{-\beta(A)}, \quad C_{\text{sieve}}(A) = \Gamma(\beta(A) + 1) \exp(\gamma\beta(A) - B(A)),$$

by a Selberg sieve at level $z = x^{1/4}(\log x)^{-B}$ with Bombieri–Vinogradov remainders (the $A = 3$ case is exactly $T(3)^+$).

A crucial supporting fact, proved as a genuine unconditional theorem (Stirling’s formula applied to the explicit bound $\beta(A) = O(\log A \log \log A)$, itself a consequence of $\varphi(n) \gg n/\log \log n$), is that the Γ -factor’s contribution to $\log C_{\text{sieve}}(A)$ is $\exp(o(\log^2 A))$ — sub-critical against the $\log^2 A$ covering law, so this apparatus cannot by itself resurrect the covering density’s own scaling. [MEASURED] the Mertens-constant sum $B(A)$ is computed at two independent cutoffs ($Q = 10^6, 10^7$) and agrees to three decimals, staying in $[0.32, 0.47]$ for $A \leq 200$; its *expected* size $O(\log A)$ follows from Norton’s lemma [41] together with the typical size of $L(1, \chi)$ in the Granville–Soundararajan circle [43].

Claim 4.6 (Lemma FL, cited). *The dimension-uniform fundamental lemma (Halberstam–Richert [39]; Friedlander–Iwaniec [40]) extends $T(A)^+$ ’s main term and remainder uniformly in A , for $A \leq \exp(c\sqrt{\log x})$, with effective constants.*

Claim 4.7 (E_lane, [CLAIMED]). *Optimising the Selberg exponent against Lemma FL’s growing-level constraint, there exist effective $c, c' > 0$ such that the number of hard primes $p \leq x$ escaping the $d = 1$ aligned-prime lane at width $A(x) = \exp(c\sqrt{\log x})$ satisfies*

$$E_{\text{lane}}(x) \ll x \exp(-c' \sqrt{\log x} \log \log x).$$

E_lane is deliberately *weaker* than Vaughan’s 1970 bound: the $d = 1$ lane carries only $\log A$ worth of sieve mass, against the full covering box’s $\log^2 A$. It is presented as the provable, effective “floor” the aligned-prime sieve alone can deliver — a useful calibration point rather than a record.

4.7 E_power: an unconditional exceptional-set bound beating Vaughan

The programme’s single strongest unconditional analytic result concerns the *full* covering box on *all* integers (not only hard primes) — recall from Section 3.4 that Lean’s Covered/Survivor predicate on the full (a, c, d) box is a *superset* of the true ES exceptional set, so an upper bound here is automatically an upper bound there.

Lemma 4.8 (Lemma SM, second moments, [CLAIMED] with a written proof). *Let $Q(A)$ be the moduli $q = 4acd - 1$ of the Lean covering cells ($1 \leq a, c \leq A$, $1 \leq d \leq 5$), $\mu(A) = \sum_q 1/q$ the total mass, $\mu_\ell(A) = \sum_{\ell|q} 1/q$ the mass divisible by a prime ℓ , $\Delta(A, T) = \sum_{\ell > T} \mu_\ell^2$, and $\delta_{\text{ub}}(A, T) = \max_q \sum_{\ell|q, \ell > T} \mu_\ell$. There is an absolute effective C such that for all $A \geq 3$ and $3 \leq T \leq 20A^2$,*

$$\Delta(A, T) \leq C \left(\frac{\mu(A)^2}{T} + \frac{(\log 2A)^2}{\sqrt{T}} + \log \log(2A^2 + 3) - \log \log T + 1 \right),$$

$$\delta_{\text{ub}}(A, T) \leq C \mu(A) \left(\frac{\log 2A}{T \log T} + \frac{1}{\sqrt{T} \log 2A} \right).$$

The proof (reproduced in full in `erdos-strauss-E-power.md`) is elementary: it reduces to a divisor-in-arithmetic-progression bound $\sum_{n \leq x, n \equiv a(q)} d(n) \ll (x/q) \log(2x) + \sqrt{x}$, a weighted-harmonic corollary, a pointwise bound on μ_ℓ , and finally a second-moment sum over primes $\ell > T$ split by Mertens’ theorem. On the schedule $T = \mu/\eta$ for fixed small $\eta \in (0, 1/4]$, where $\mu(A) \asymp (\log A)^2$, both error terms in Suen’s inequality (Section 4.1) become $\Delta e^{2\delta} = O(\eta\mu)$ — exactly the input the density argument needs. [MEASURED] the implied constants in Lemma SM are checked computationally on the Lean covering cells at A from 24 up to 2000 (`e_power_suen_moments.py`, `e_power_suen_moments_large.py`): along the schedule $T/\mu \approx 5$, the ratio $R_\Delta = T \sum_{\ell > T} \mu_\ell^2 / \mu^2$ settles at ≈ 0.13 and the Suen loss $\Delta e^{2\delta} / \mu$ stays below 0.15 throughout this two-order-of-magnitude range.

Theorem 4.9 (E_power, [CLAIMED]). *There exist effective constants $c, \delta, X_0 > 0$ such that for all $x \geq X_0$ and $A = \exp(c\sqrt{\log x})$,*

$$S_A(x, 2x) \ll x^{1-\delta},$$

where $S_A(x, 2x)$ is the number of $n \in [x, 2x)$ surviving the level- A covering box. Consequently the number of $n \in [x, 2x)$ for which $4/n$ is not a sum of three unit fractions is $\ll x^{1-\delta}$.

This is proved by combining Lemma SM’s second-moment input with Suen’s inequality (giving density $u(A) \leq \exp(-(1 - O(\eta))\mu(A))$) and a “Lemma Transfer” step: at $A = \exp(c\sqrt{\log x})$, the hub modulus $M_T = \prod_{p \leq T} p$ has size $x^{\alpha/\eta + o(1)}$ (with $\alpha = \kappa_{\text{cov}} c^2$), and for c small enough that $\alpha/\eta < 1/2$ the main term dominates the discrepancy over residues modulo M_T , giving the stated power saving. Explicitly, $\delta = (1 - O(\eta))\kappa_{\text{cov}} c^2$ for the admissible range of (c, η) ; the numerical value of δ is not the content, effectivity and positivity are. The honest comparison recorded throughout the archive: Theorem 4.9 *strictly improves* Vaughan’s 1970 exponential-saving bound $\ll x \exp(-c(\log x)^{2/3})$ to a genuine power saving, on the same object (all integers $n \leq x$, not merely hard primes) — this is, on the programme’s own account, the single strongest unconditional new mathematical statement to come out of the archive.

4.8 Gate A: the discipline separating a written record from a formalised claim

The programme applies an explicit editorial control, “Gate A” (`erdos-strauss-E-partial.md`), governing what may and may not be compiled into the Lean interface `AnalyticSurvivorBound`. Gate A *admits*: the elementary Layer-A covering machinery (already compiled); a written paper of $T(A)^+/\text{E_lane}/\text{E_power}$, with both “die conditions” of Section 4.2 stated in the abstract; the Granville–Soundararajan restatement as a dictionary into a published literature, not as extra independent characters. Gate A *forbids*: discharging `AnalyticSurvivorBound` in Lean, or any lemma whose conclusion is zero large hard survivors; treating `E_power` or `E_partial` as `H_ES` or as an assault on it; retuning κ from the flat prime-aligned Euler factor $C_{\text{euler}} \approx 0.91$ ([MEASURED], computed in `c4_euler_factor.py` to be flat in A from $A = 40$ to $A = 200$, with zero residue collisions verified for $A \leq 200$); densifying covering while the *conditional retention probability* $\text{cond}(a) = \Pr[R_a \mid R_1, \dots, R_{a-1}]$ sits at ≈ 1 ([MEASURED] at $x = 10^9$, $\text{cond}(a) \rightarrow 1$ past $a = 80$, mean 0.993 on $a = 161$ – 200 — meaning essentially all of the extra covering width beyond $A \approx 80$ contributes what the archive calls *dummy covering mass*, not genuine additional constraint); and running further large-scale computation beyond $x = 10^9$. This discipline is the explicit mechanism by which the archive prevents a genuine, useful, unconditional record (Theorem 4.9) from being silently over-claimed as progress toward the load-bearing interface.

4.9 The Granville–Soundararajan reformulation

A final analytic-lane document (`erdos-strauss-gs-reformulation.md`) restates the $T(A)$ survivor-density target in the language of pretentious multiplicative number theory: each covering condition corresponds, via quadratic reciprocity, to a real Dirichlet character condition, so the joint survival event is a large-deviation lower tail of a sum of quadratic character values at primes — precisely the object the Granville–Soundararajan pretentiousness programme [44, 45] and Lamzouri’s large-deviation theorems for character sums were built to compute. The document is explicit that it contains *no new theorem*; its role is to name the correct published dialect for the open question of Section 4.10, and it records two structural caveats: the number of covering moduli in play ($A = \exp(c\sqrt{\log x})$) is small enough that a naive application of Heath-Brown’s quadratic large sieve [46] is trivial, so the genuine leverage would have to come from *growing moments* rather than a raw second-moment bound; and character-sum cancellation, unlike interval-intrinsic randomness, is *not* capped by the k -budget invariant of Section 4.1, so this is one of the few investigated directions

the invariant does not pre-refute. A companion measurement, the “C2” cross-moduli Jacobi-symbol correlation test (`c4_c2_symbols.py`), checks directly whether reciprocity among the forced covering symbols produces *super*-multiplicative shrinkage beyond the deterministic exclusions already counted; [MEASURED] at $x = 10^9$, $A \leq 80$ it does not: mean $|z|$ on covering-modulus pairs (0.73) is if anything *smaller* than on inert pairs (0.89), and the sign of any residual bias is wrong for it to explain the observed selection constant $\widehat{C}(80) = 2.69$.

4.10 The analytic workload: an open question posed to the sieve literature

The programme maintains one further document, `erdos-strauss-sieve-desk.md`, whose own header describes it as “the analytic question list. It is not a theorem, not a request to prove the Erdős–Straus conjecture, and not a request to densify covering.” Unlike the working notes discussed above, it is written to be legible to a reader outside the programme: it restates the $T(A)^+/E_lane/E_power$ results as theorems already established in this archive, restates the $T(3)$ completion question and the negative findings of Sections 4.4 and 4.5 as the open remainder, and names the relevant published dialect as the half-dimensional/ vector-sieve literature on ker-character conditions at primes — in particular the active $m^2 + n^2 + 1$ literature (Hooley [53]; Iwaniec [47, 48]; Nath–Xie [66]; Fuchs–Hsu–Rickards–Schindler–Stange [67]). It is explicit that this is a bounded mathematical question, not a request for anyone to prove ES, and that it should not be mixed in the same discussion with the geometric question of Section 5.10: the two are addressed to different published literatures and share only the identification, discussed in Section 5.9, that both ultimately reduce to the same analytic core.

5 Route 2: the Bright–Loughran geometric approach

5.1 Background: the Brauer–Manin theory of Erdős–Straus surfaces

Bright and Loughran [13] study, for each $n \geq 2$, the affine cubic surface $U_n : 4xyz = n(xy + yz + zx)$ over $\mathbb{Q}$, showing $\text{Br } U_n / \text{Br } \mathbb{Q} \cong \mathbb{Z}/2$, generated by the transcendental quaternion algebra $\alpha = (-u_1/u_3, -u_2/u_3)$, computed on the desingularisation $\widetilde{U}_n$ (whose complement of three boundary lines is a torus $\mathbb{G}_m^2$). Their principal results, all formalised or directly used in this archive:

- **Theorem 1.2:** every natural-number solution satisfies $\prod_{p|n} (-u_1/u_3, -u_2/u_3)_p = -1$ (Section 3.5);
- **Theorem 1.5:** non-natural integer solutions give +1 instead (Section 3.7, the octant invariant);
- **Theorem 1.8:** there is *no* Brauer–Manin obstruction to ES itself — the conjecture survives their computation, but this is a non-obstruction statement, not an existence proof;
- **Theorem 1.9:** Brauer–Manin is *not* the only obstruction to strong approximation on these surfaces — the exhibited failure is a consequence of the *finiteness* of integral points (the box, Lemma 3.10) via Lang–Weil, i.e. a density phenomenon, not an obstruction-theoretic one.

The programme’s own summary of the open mathematical content, after formalising this apparatus: *ES is the question of occupancy of one real component (C_+ , the positive octant) of a surface that already has integral points in the other* (mixed-sign points always exist, `esZ_nonempty`).

5.2 The formalised $\mathbb{Z}$ -model and TUB-EP

Section 3.7 already describes the Lean content in detail: the affine $\mathbb{Z}$ -model, the box bound, the octant invariant, and the central named hypothesis `TubEpHypothesis` — occupancy of the positive

octant for every $n \geq 2$ — proved *equivalent* to `ErdosStraus` itself (`tub_ep_iff_erdos_straus`). This equivalence is treated throughout the archive as a methodologically important negative fact rather than a positive step: requirement R2 of the acceptance specification (Section 8) demands that a genuine load-bearing conjecture C have mathematical content *beyond* ES itself, and a certified equivalence is explicitly disqualified as “a translation, not a target” with “zero strategic distance”. The geometric route’s honest status, reached only after the negative investigations below, is that it supplies *class-theory and an effectivity framework* — a precise vocabulary in which a well-posed question about a class of surfaces can be stated — rather than a proof mechanism with tools that presently exist.

5.3 Harpaz’s descent-fibration theorem: inapplicable, and provably so

The first candidate geometric mechanism tested is Harpaz’s theorem on integral points of conic log K3 surfaces [17], whose engine is Swinnerton-Dyer-style descent on norm-1 tori of a quadratic field, contingent on independence, in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, of a specific finite collection of classes attached to the fibration. As established unconditionally in Section 3.8 (`ConicFiber.fiber_identity`), *every* conic fibre of U_n , for every $n \geq 2$ and every choice of fibre parameter t , is algebraically split: $uv = (nt)^2$. On a split conic there is no non-trivial Selmer structure to compare, and Harpaz’s hypotheses fail degenerately rather than merely being hard to verify. The programme records this both as a negative result for the naive application, and, via `fiber_to_divisor/divisor_to_fiber`, as a positive structural theorem: along every fibration, TUB-EP’s existence question *is* the divisor problem of Section 2 — the two routes are, at the level of this one surface family, literally the same question in two coordinate systems.

5.4 Cao–Xu toric strong approximation: the wrong theorem

A second natural candidate is Cao–Xu’s theorem that Brauer–Manin suffices for strong approximation on genuinely toric varieties [20]. The desingularised surface $\tilde{U}_n$ does contain a torus $\mathbb{G}_m^2$ in its interior, but — as recorded in BL’s own Remark 3.12 and reconfirmed in the plan’s route-2 dossier — the natural torus action does *not extend* to the boundary compactification, so $\tilde{U}_n$ is not an honest toric variety in the sense Cao–Xu’s theorem requires. `[LEAN] esQ_pos`: the existence of a positive *rational* point $(n, n, n/2)$ independently of any BM computation is recorded as a direct illustration of why Cao–Xu cannot, by itself, be what fills the octant with an *integral* point.

5.5 Candidate C5 and its refutation chain: the real-compatible twist

The most substantial single geometric idea investigated is candidate conjecture C5: since the Brauer class α is exactly what obstructs strong approximation, perhaps descending through α — replacing U_n by a torsor on which α trivialises — produces *non-split* conic fibrations to which Harpaz’s engine genuinely applies. The investigation, entirely documented and largely kernel-checked, proceeds through three stages:

1. **C5 (naive, refuted same day).** The naive α -trivialising cover $w^2 = -u_1u_3$ is shown, by Section 3.7’s octant invariant, to have *no real points at all* over the positive octant (since $u_1u_3 > 0$ there), formalised as `[LEAN] TwistDescent.naive_cover_empty`. The Brauer class does not merely obstruct existence; it walls its own trivialising cover off from the octant entirely.
2. **C5₁ (corrected, examined and re-scored).** The natural repair replaces the trivialising cover by *real-compatible twists* $w^2 = d \cdot u_1u_3$ ($d > 0$ squarefree), which do have real points over C_+ by construction. Every positive solution of U_n is shown to lift to some twist (`TwistDescent.lift`, Gate 2), with $d = O(n^2)$ forced by the box bound — ruling out any fixed, n -independent twist

family (which would fail requirement R4, Section 8). A hand calculation shows the pulled-back quaternion data on the twist takes the shape $(-d, -u_2u_3)$; fibering by $u_2 = t$ and rationalising, integrality on the twist is shown to be equivalent to a divisor condition on norm-form values over $\mathbb{Q}(\sqrt{-d})$ — a genuine toolkit gain (imaginary-quadratic factorisation and Chebotarev machinery become available), but also, honestly, a restatement rather than an evasion of the same analytic core.

3. **C5₂ (Gate 3, negative, closed).** The specific t -fibration named in C5₁’s derivation is shown to possess an *explicit rational section* over $\mathbb{Q}(t)$ (Section 3.9, `t_fiber_section_cleared`) — the generic fibre is *split for every d* , reproducing exactly the same obstruction one level up the tower. Harpaz’s Theorem 1.0.1 (the specific, unconditional case of his theorem applicable here, whose hypotheses require non-split fibres and whose conclusion is in any case only S -integral with $2 \in S$, not a genuine $\mathbb{Z}$ -point of C_+) does not apply. A same-day autopsy (`u1_fiber_section_cleared`) shows the identical rational curve, merely reparametrised, is *also* a section of the projections $\pi = u_1$ and $\pi = w$ — changing the linear projection does not open a genuinely non-split fibration of this surface.

The programme records an explicit methodological lesson from this chain: *run the section test first* — before any Selmer or descent architecture is discussed for a proposed fibration of any surface in this family, either exhibit a $\mathbb{Q}(t)$ -section (which closes the route immediately) or prove non-splitness of the generic fibre (which opens it). The line is recorded as closed pending a genuinely new, non-split fibration, none having been found.

5.6 Candidate C3 and its freeze: Campana points do not reach C_+

Candidate C3 proposes a triangle-boundary semi-integral (Campana/orbifold) existence principle, motivated by the observation that ES pairs have a totally split boundary triangle with interior $\mathbb{G}_m^2$. Three checks (recorded as C3₁) freeze this line:

1. ES pairs *do* have the claimed geometric structure (a paper fact BL already use, not a new theorem);
2. Campana/semi-integral points do *not*, in general, imply a genuine $\mathbb{Z}$ -point of C_+ : Mitankin–Uhlemann [26] prove that Markoff orbifold pairs satisfy the semi-integral Hasse principle while often having *empty* integral point sets — their own Remark 1.5 states explicitly that Campana/Darmon points “do not behave in the way that $\mathcal{U}_m(\mathbb{Z})$ does”; consequently a class-level Campana Hasse principle, even with empty obstruction, does not bridge to **TubeHypothesis**;
3. there is no named theorem T in the literature that supplies effective existence of a $\mathbb{Z}$ -point of C_+ on the ES pairs by this route.

A further claimed discriminator — that the Markoff boundary is an irreducible conic rather than a triangle, and so C3’s triangle hypothesis would automatically exclude the Markoff counterexample — is checked directly against Mitankin–Uhlemann’s actual compactification and found to be *false*: their model $x_0(x_1^2 + x_2^2 + x_3^2) - x_1x_2x_3 = mx_0^3$ has boundary $D = \bigcup_i \{x_0 = x_i = 0\}$, itself a triangle of three lines. The triangle hypothesis does not exclude the known negative comparison case.

5.7 The Schinzel separator programme

To test what an occupancy-forcing principle would need beyond mixed-sign existence and the box, the programme studies the generalised Schinzel family $m/n = 1/x + 1/y + 1/z$ directly. Section 3.10 records the Lean content; the mathematical narrative is:

- An infinite family $(2z - 1)/z$ ($z \notin \{3, 4, 6\}$) has an explicit mixed-sign point but no positive solution, since positive solvability would require $(z - 2) \mid 4$. The flagship case $z = 5$ is $9/5$.
- **[MEASURED]** A bounded computer sweep of the “narrow band” $1 < m/n < 3$, $n \leq 40$, finds **86 separators** in lowest terms (mixed-sign witness exists, positive search provably empty — the positive search is complete for $x \leq 3n/m$) (`s1_schinzel_separators.py`); only the $9/5$ and $13/7$ instances are Lean-verified (Section 3.10), the remaining 84 being script-level results.
- **The mechanism, identified precisely.** For m/n bounded strictly between 1 and 3, the positive search tree has $O(1)$ branching (the leading denominator is bounded by $3n/m = O(1)$), so failure is a *finite, Archimedean-integral* phenomenon — Bright and Loughran’s own Theorem 1.9 finiteness mechanism, promoted from an obstruction to strong approximation into an outright existence failure. Ordinary Brauer–Manin machinery, which is not sensitive to Archimedean tree size, does not detect this.
- **The regime split ($S1_1$).** The Schinzel class splits into a *narrow* band (m/n bounded away from 0), where the tree is finite and existence reduces to a decidable computation (`decideCplus`, Section 3.10), and a *wide* band ($m/n \rightarrow 0$, tree width $\sim n/2$ for $m = 4$) — which is exactly where Erdős–Straus itself lives. A “wide-tree separator” cannot, by construction, occur in the narrow band; extending the 86-pair sweep therefore cannot test whether any occupancy principle is *sufficient* in the regime that matters.
- A hand (non-kernel) adelic sketch on $9/5$ tests whether α obstructs a C_+ -restricted integral adele there: two explicit points $((1, -5, 1)$ at good places, $(9/11, 45/11, 3)$ as a $\mathbb{Z}_2$ -point) are assembled with hand-computed Hilbert symbols at 2, giving product of invariants $+1$ — i.e. *no* obstruction is found on this sketch, meaning the $9/5$ separator’s failure is invisible to the Brauer–Manin machinery precisely as the finite-tree mechanism above predicts. This sketch is explicitly labelled non-kernel and is presented as a computation to be confirmed by specialists, not a result.

Claim 5.1 (Candidate C11, refined existence schema — not proved). *Mixed-sign occupancy + the box + no integral Brauer–Manin obstruction against C_+ + Archimedean non-degeneracy (the admissible leading-denominator range growing with n , equivalently $m/n \rightarrow 0$ along the family) implies positive-octant existence.*

C11 is scored honestly against the eight-point acceptance specification in Section 8 and found to satisfy none of the eight outright; its role in the archive is as the sharpest currently stated version of what a genuine geometric existence principle for this family would have to assert, together with the explicit necessity proof (the 86 separators) that its fourth hypothesis cannot be dropped.

5.8 No Vieta: why the Markoff/Bourgain–Gamburd–Sarnak playbook cannot be imported

Section 3.11 records the Lean content of the final structural finding: the ES/Schinzel affine equation is linear, not quadratic, in each coordinate, so it admits no second root and hence no Vieta-type involution analogous to the Markoff surface’s $x \mapsto 3yz - x$. This is the formal reason the entire Bourgain–Gamburd–Sarnak strong-approximation machinery [22, 23] for Markoff-type surfaces — the natural comparison class, since $U_n(\mathbb{Z})$ is never empty exactly where sibling Markoff surfaces frequently have *no* integral points at all [24, 25] — has no orbit structure to act on here.

5.9 Synthesis: the two lanes are one lane

The geometric investigation’s terminal finding (`erdos-strauss-novel-structures-plan.md` §4v, 20 August 2026 evening) is that every geometric mechanism considered reduces to the same analytic core:

- fibrations reduce to divisor conditions (Section 3.8), conserved under the real-compatible twists (Section 5.5);
- the universal torsor of $\widetilde{U}_n$ is literally the witness equation, so torsor-counting (candidate C10) is the Elsholtz–Tao count [31], whose all- n lower bound is again the divisor-dispersion problem of Section 4;
- the affine equation’s linearity (Section 3.11) rules out any Vieta/Markoff dynamical mechanism, closing that branch of the inventory entirely.

The consequence drawn is not that geometry is worthless, but that its honest role is **class-theory and an effectivity framework**: precise, publishable foundational questions (Section 5.10) about a class of strongly-obstructed log K3 surfaces, with an explicit effectivity framework as a genuine open problem in its own right — not a currently available QED engine. Both lanes converge on the same object: for every real geometric mechanism tried, the residual open mathematics is a question about the joint distribution or divisor structure of shifted values, exactly the object $T(A)/T(3)$ already name analytically.

5.10 Open geometric questions in the Bright–Loughran literature

The programme records five bounded, numbered open questions in a document, `erdos-strauss-loughran-orbit.md`, whose own header describes it as “the geometric worklist. It is not a theorem, not a request to prove the Erdős–Straus conjecture.” The document identifies the relevant literature as the body of work descending from Bright and Loughran’s founding paper [13] — Mitankin, Uhlemann, Colliot-Thélène, Wei, Xu, Harpaz, Uppal, Jahnel, and Schindler among others — and states the five questions purely as mathematics, without addressing them to any named individual:

1. whether the notation $(U(\mathbb{A}_{\mathbb{Z}})^{C+})^{\text{Br}} := (C_+ \times \prod_p U(\mathbb{Z}_p))^{\text{Br}}$ is the right formalisation of “no integral Brauer–Manin obstruction against a real component”, and whether Theorem 1.8 amounts to its nonemptiness for the ES pairs;
2. whether the two 2-adic Hilbert symbols in the non-kernel 9/5 adelic sketch of Section 5.7 are correct, and whether the construction is the correct analogue of Theorem 1.8 on the 9/5 Schinzel model;
3. whether any refined obstruction in the semi-integral literature detects “the positive search tree is finite and empty” — the mechanism the Schinzel separators exhibit but which ordinary Brauer–Manin does not see;
4. whether any class-level framework in this literature asserts anything in the *wide* regime ($m/n \rightarrow 0$) — the regime in which Erdős–Straus itself lives, as opposed to the narrow band in which the 86 Schinzel separators were found; and
5. whether TUB-EP, or the refined schema C11 (Claim 5.1), is a statement that literature would recognise, reject, or refine as a genuine class principle.

The document is explicit about what these questions are *not*: not a request to apply Harpaz, Cao–Xu, or Theorem 1.8 (each already shown inapplicable or insufficient, Sections 5.3 and 7.1); not a request to explain why ES models have integral points when Markoff models do not (the premise conflates two different statements, Section 3.11); and not a request to densify covering or to treat any external claimed proof as settled (Section 6). A companion table in the source document lists further, more specialised literatures (Jahnel–Schindler and Colliot-Thélène–Wittenberg on affine cubics; Mitankin–Uhlemann on Campana points; the dispersion/DFI literature behind Section 4.10; Schinzel’s own polynomial-identity literature) as adjacent questions to be kept separate from the five above and from the analytic question of Section 4.10.

6 External claims and the prior archive

Alongside its own investigation, the programme carries out two triage exercises against material external to this archive: an examination of a sibling, separately maintained research effort (referred to throughout as the “prior Track-1 archive”), and a direct check of a 2026 preprint claiming an elementary proof of the conjecture. Both are documented explicitly as *library/documentation*, *not a merge*, and neither changes the status of any result reported in Sections 4 and 5.

6.1 The prior Track-1 archive: what was kept, and what was not

`erdos-strauss-prior-archive.md` documents a sibling repository — not part of this archive — pursuing a different existence-assault strategy: a “gateway” construction asserting that every hard residual prime admits a bound $A \leq C$ in $\mathbb{F}_A^\times$ (a working constant $C = 500$, with an empirical maximum $A = 211$ observed through 6×10^9), packaged as finite-catalogue-type (FCT) and gateway covering across hundreds of generated Lean files. The document is explicit about the verdict: *“This programme already classified that vehicle as interval-intrinsic covering densification. Extra slices do not empty a QED-scale box”* — i.e. the sibling effort is read as another instance of the density-versus-universality pattern this programme’s own k-budget invariant (Section 4.1) already explains. Table 1 summarises what was, and was not, brought into this archive as a result.

Artefact	Kept here as	Role
Størmer’s theorem on consecutive smooth numbers (Apache 2.0; Mathlib PR #42040)	<code>Stormer.lean</code>	Library theorem only; not on the $T(A)$ /geometry path (Section 3.13).
Bello-Hernández-Benito-Fernández Lemma 3/Theorem 8, and an examination of Bradford 2026	<code>erdos-strauss-bradford-lemma3.md</code>	Documentation of a finite-parameter covering no-go, and of why Bradford 2026 does not establish its covering claim (Section 6.2).
Leochlon Type-II identity $(4b - 1)(4c - 1) = 4p\delta + 1$	<code>erdos-strauss-leochlon.md</code> , <code>Leochlon.lean</code>	Dictionary into <code>ES.IsES</code> (Section 3.12); not an FCT/gateway merge.

Table 1: What the prior Track-1 archive contributed to this repository.

Explicitly *not* brought into this archive, per the same document: the generated covering cascade (`Cover*`, `Hit*`, `Band*` Lean files); the `HasGatewayNsqWitness` bounded- A construction as a completeness vehicle; a Paley–Zygmund-based ensemble argument proposed as an every- n proof; a claimed Matthiesen–Wang interface discharge of a “ForcedSix” obstruction (the prior archive’s own record already states that it does not discharge it); further finite-catalogue growth; and any zkVM/SP1 verified-execution wrapping. The document’s closing instruction is unambiguous: do

not treat `Stormer.lean` or `Leochlon.lean` as progress toward `erdos_straus_of_interface`, and do not reopen the sibling archive’s Bounded-*A* construction or Bradford 2026 as a proof.

6.2 Bradford 2026 and the Bello-Hernández–Benito–Fernández rigidity lemma

`erdos-straus-bradford-lemma3.md` examines two related pieces of external literature directly.

Lemma 3 / Theorem 8 (Bello-Hernández–Benito–Fernández). The statement, quoted from [12] Theorem 8 with the proof deferred to [79] Lemma 3, is a rigidity result: one cannot generate a finite number of congruence classes containing *all* primes $p \equiv 1 \pmod{4}$ by fixing, in the classical Type I/Type II witness equations, three of the four parameters to a finite set and leaving the fourth free. The proof idea is elementary lcm arithmetic — each frozen triple produces an arithmetic progression or finite set of n ; taking the lcm T of the moduli attached to any finite collection S of triples, the progression $\{4Tt + 1 : t > 0\}$ cannot arise from any triple in S , since matching the parametric formula would force 1 itself to be an Erdős–Straus number. The document records precisely what this does and does not rule out: it rules out any *finite* catalogue of fixed-parameter identities, matching this programme’s own “do not densify covering” discipline exactly; it does *not* rule out constructions whose free parameters grow with n (there being then no single finite S and no single modulus T) — such constructions merely lie outside the lemma’s hypothesis, which is not the same as a confirmed density escape.

Claim 6.1 (Bradford 2026, [NEGATIVE RESULT]). *K. Bradford’s preprint [80] claims an elementary proof of ES for every prime, treating $p \equiv 1 \pmod{4}$ via Type I/Type II identities and asserting that combining them yields a covering system. The programme’s examination finds the covering claim unsubstantiated: the preprint’s only written verification is a $k = 0$ list of four moduli (44, 20, 8, 140) and the observation that 5, 13, 17, 29 are the first four primes $\equiv 1 \pmod{4}$; there is no verification that every such prime lies in one of the listed progressions, no table for any $k \geq 1$, and no engagement with Mordell’s quadratic-residue obstruction. Three specific reasons the covering claim fails are recorded: (i) Mordell’s classical obstruction (Section 1.2) already forbids any finite family of polynomial identities from covering $p \equiv 1 \pmod{840}$, since 1 is always a square, and Bradford’s Lemmas 1–2 are identities of exactly this type; (ii) the written proof exhibits only $k = 0$, so it is precisely the finite-parameter cover that Lemma 3 above rules out, and no infinite template over all k is supplied; (iii) listing four small primes is not a covering argument; and (iv) several of the listed arithmetic progressions (13 mod 20, 29 mod 44) coincide with identities already present in the classical Mordell layer of this archive’s own Lean development (Section 3.4.2).*

The document’s conclusion is that formalising Bradford’s claim as a proof is not worthwhile, and that formalising its Lemmas 1–2 as general identity schemas would itself constitute the covering densification this programme deliberately avoids; consequently neither is carried into Lean. Three further Bradford-adjacent publications [81, 82, 83] are noted as distinct papers that do not remove the Mordell obstruction either.

7 The candidate-conjecture catalogue

`erdos-straus-candidate-conjectures.md` maintains a running list of named candidates for the load-bearing statement C that a proof of ES would ultimately rest on, each scored against the acceptance specification of Section 8 and each carrying an explicit statement of its *primary risk* — the requirement most likely to kill it. The document’s own framing is important: “None is asserted true; several are expected to fail. That is what a candidate list is for.” Section 7 summarises the full

list and its current status; candidates discussed in detail elsewhere in this report are cross-referenced rather than re-derived.

ID	Statement (informal)	Input class	Status / primary risk
C1	Structured-modulus dispersion: level of distribution 1 for the joint divisor condition, uniformly over the growing (a, d) family	spectral/bilinear	Reference candidate; “H_ES in its most attackable dress”. Risk R2: may be ES relocated.
C2	Reciprocity coupling: forced character conditions compound <i>beyond</i> their raw covering mass via quadratic reciprocity	algebraic rigidity	[MEASURED] Does not fire (Section 4.9): cross-moduli Jacobi-symbol correlations are noise, wrong sign for $\hat{C} > 1$.
C3	Triangle-boundary semi-integral (Campana) existence principle	global geometry	Frozen at C3₁ (Section 7.1): Campana points do not reach C_+ ; the claimed Markoff-excluding discriminator is false.
C4 / C4 ₁	Simultaneous shift-roughness / T(A) : joint density of class-roughness at fixed covering width	spectral (Wirsing/LSD) + compositum densities	The programme’s primary in-house line (Section 4.2); T(3) opening case claimed at the upper-bound half (Section 4.3).
C5 / C5 ₁ / C5 ₂	Descent through α : real-compatible twist $V_{n,d}$	global geometry (Harpaz-type)	Refuted / closed (Section 5.5): the named t -fibration is split (Gate 3, kernel-checked); “run the section test first”.
C6	Theory-4 flatness: Selberg- L^2 residual of the survivor indicator is ε_j -flat against later covering shells	spectral (new equidistribution)	Rungs 1–3b partially executed (Section 4.1 discussion); primary risk R2/R3, may be H_ES relocated at asymptotic scale.
C7 / C7 ₁	Bounded-coupling covering family: a new identity family breaking the Poisson coupling law	none (combinatorial discovery)	[MEASURED] First search negative (Section 9.4): 193 degree-1 identities searched, none covers a hard class; bounded- ω mass does not break the coupling law at toy scale. Probably false.
C8	Thin-exception mop-up: a two-stage Frobenius-thin containment plus effective Chebotarev closure	spectral + rigidity	Not pursued computationally; primary risk is the classical “for-all gap” at R4 (thinness historically gives density zero, not emptiness).
C9	Discriminant-uniform spectral equidistribution (theta route): a p -dependent uniform DFI-type statement	spectral (GL(2)/metaplectic)	Not pursued; primary risk is that p -dependent uniformity at this level is beyond current spectral technology.
C10	Effective uniform log-Manin on the ES Cox ring (universal torsor counting)	mixed geometry/circle method	Identified as C1/C4 in disguise (Section 5.9): the torsor is the witness equation, so this is the Elsholtz–Tao count.
C11	TUB-EP-refined: mixed-sign + box + no integral BM obstruction against C_+ + Archimedean non-degeneracy $\Rightarrow$ positive existence	global geometry (schema)	Claim 5.1; the sharpest current geometric schema, satisfies none of R1–R8 outright; necessity of the fourth hypothesis proved via the 86 Schinzel separators.
NV	No Vieta (closes the geometric mechanism inventory)	–	[LEAN] Proved (Section 3.11): the affine equation is linear in each coordinate; no second root, no Markoff-type involution.

ID	Statement (informal)	Input class	Status / primary risk
S1 / S1 ₁	Schinzel separator suite; narrow/wide regime split	–	Section 5.7: 86 separators found; the wide band is where ES itself lives; a non-kernel adelic sketch on 9/5.

7.1 C3, restated for completeness

See Section 5 for the full discussion of C3/C3₁, C5/C5₁/C5₂, S1/S1₁, C11 and NV, and Section 4 for C1/C2/C4/C4₁/C6/C10. The programme’s own portfolio note groups C1, C4, C4₁, C6, C9 and C10 as “one object”: they die together if the selection constant $C(A)$ turns out to be super-critical in $\log^2 A$ ($c' \geq \kappa$), or if the conditional retention probability $\text{cond}(a)$ sticks at 1 (dummy covering) out to QED scale; as measured (Section 4.2, Section 4.8), growing A to 200 at $x = 10^9$ showed neither has yet happened, but the margin against κ is thin ($c' = 0.104$ against $\kappa = 0.139$) and the dummy-covering trend is the more immediate concern.

8 The acceptance specification

`erdos-strauss-conjecture-spec.md` formulates, as an explicit deliverable in its own right, an *acceptance specification* for what any candidate load-bearing conjecture C must satisfy in order to genuinely solve ES rather than merely restate it. The document frames this as itself a piece of mathematics: “Each requirement is traceable to a theorem, measurement, or refutation in the programme record; each has already eliminated at least one proposed candidate.” The eight requirements, with the notation A for the covering level, $\kappa \approx 0.139$ for the measured marginal constant, and $q \mid p + 4a^2d$, $q \equiv -1 \pmod{4ad}$ for the divisor form:

R1 (Verified sufficiency). C must imply `HardLandingHypothesis` — hence ES — through kernel-checked implications. *Status:* *cheap and fully under the programme’s control*; both named interfaces satisfy it by construction (Section 3.4.4).

R2 (Independent content). C must be attackable by methods that do not already amount to proving ES; a certified equivalence is a translation, not a target. *Status:* *the open risk for essentially every candidate* — `TubEpHypothesis` fails it outright (Section 3.7); $T(A)/C1$ ’s relocation risk is flagged as needing an early expert check.

R3 (Barrier evasion — the k-budget invariant). C ’s proof must inject arithmetic input from *outside* the interval-intrinsic class capped by Section 4.1: admissible classes are spectral/bilinear arithmetic, algebraic rigidity used non-statistically, or genuinely global geometry not routed through fibres. *Any C whose natural proof is “sieve harder” fails R3 before it starts.*

R4 (Exceptional-set annihilation, with mandatory growth). Fixed parameter boxes provably leave $\sim x/(\log x)^{1+c(D_0)}$ exceptions (compounding failure densities across a fixed box); C must involve a level growing with x at rate $\log^2(\text{level}) \gtrsim \log x$, i.e. $\text{level} = \exp(c\sqrt{\log x})$ with $\kappa c^2 > 1$. *Status:* *eliminates every fixed-box proposal outright*, including an externally advised fixed- D_0 landing lemma (shown heuristically false, Section 4.1 discussion of plan §4q).

R5 (Splitness compatibility). C must locate the arithmetic in integrality and divisors, not in fibre-level local–global structure, since every conic fibre of the ES surface is split (Section 3.8). *Status:* *satisfied natively* by the divisor form (Section 2); **fails** for every fibration-based geometric mechanism tried (Section 5.5).

R6 (Consistency with known negatives). C must not prove too much: it must be compatible with genuine Markoff-type integral Hasse failures [23, 24, 25], with BL Theorem 1.9, and with abelian insufficiency (no finite set of congruence conditions covers the hard classes). *Status:* eliminates C3 as originally stated (Section 7.1).

R7 (Effectivity). C must come with an explicit threshold X_0 — an ineffective Siegel-style existence statement would establish ES as true while leaving the machine-checked proof permanently uncompileable. *Status:* satisfied by the finite certificate mechanism of Section 3.4 in principle; open for every geometric candidate.

R8 (Statability now, in a published dialect). C must be formalisable as a kernel **Prop** today, and phrased in a published dialect — bilinear forms in divisor problems, or global existence principles for log K3 surfaces — so that the statement has existing theorems to cite, not only ES. *Status:* both named interfaces satisfy (a); the analytic worklist (Section 4.10) and the geometric questions (Section 5.10) are the programme’s record of (b).

Table 3 reproduces the specification document’s own scorecard for the two live candidates it regards as closest to satisfying all eight requirements simultaneously.

Requirement	DivisorLandingBound bound)	(growing	Existence principle $\Rightarrow$ TUB-EP
R1 sufficiency	proved		via <code>erdos_straus_of_tub_ep</code> once C is stated
R2 independence	open — the known risk		open — if stated for a genuine surface class
R3 barrier evasion	designed for spectral input; unproven		automatic (not interval-bound)
R4 growth	built in		must be engineered (uniformity in n)
R5 splitness	native (it is the divisor form)		the hard design constraint
R6 negatives	quantitative, not local–global		the hard design constraint
R7 effectivity	plausible (dispersion proofs usually effective)		needs care
R8 statability/dialect	stated; dispersion school		statable schema; Bright–Loughran literature

Table 3: The acceptance-specification scorecard, reproduced from `erdos-strauss-conjecture-spec.md`. Neither candidate currently satisfies all eight requirements.

The specification’s own one-paragraph summary captures the programme’s central methodological stance precisely enough to reproduce verbatim: “*A conjecture that solves Erdős–Straus must be a kernel-statable proposition with mathematical life beyond ES, whose proof injects spectral, rigidity, or global-geometric input outside the interval-intrinsic class, which annihilates (not merely thins) the exceptional set through a parameter growing like $\exp(c\sqrt{\log x})$ with $\kappa c^2 > 1$, which respects the surface’s total splitness by living in divisor/integrality structure or in genuinely global geometry, which is consistent with the Markoff failures, Theorem 1.9, and abelian insufficiency, which names its threshold, and which is written in a published dialect with existing theorems, not only ES. Everything short of this is infrastructure — valuable, publishable, and already substantially built — but not the keystone.*” The specification document’s own closing attribution, which this report reproduces in place of an earlier, superseded version naming specific model systems, reads: “*Prepared as part of the Riley Betts Ltd Erdős–Straus programme, facilitated by Martyn Riley (automation and high-level strategy; not a claim of expertise in number theory, combinatorics, or geometry). The mathematical*

work is a collaboration of Anthropic agents, Grok 4.6, and Cursor agents, with Lean 4, Mathlib, and numerical test resources. No proof of the conjecture is claimed. Every claim above is traceable to a kernel-checked theorem, a labelled measurement, or a recorded refutation in the programme archive.”

9 Computational investigations

Every quantitative claim in Sections 4 and 5 is backed by one of twenty Python scripts in the archive, each producing a JSON artefact recording its parameters and results. This section describes each script’s purpose and principal findings; Appendix A additionally lists every script and its output file(s) for reference. All computation is capped, by explicit programme discipline (“Do not run $x = 10^{10}$ ”), at $x \leq 10^9$; this ceiling is stated repeatedly and is treated as a hard methodological boundary rather than a resource constraint — the archive is explicit that further scans at the same scale would refine constants without resolving the open die-conditions.

9.1 The T(A)/ClassRough family

Twelve scripts instrument the T(A) investigation of Section 4.2.

c4_S_xscan.py The primary x -scan of $S(A, x)$ at fixed $A \in \{40, 80\}$ across $x = 10^5$ to 10^9 (1,587,581 hard primes at the top scale). Provides the sieve of hard primes and the **ClassRough**-equivalent Boolean predicate reused by most later scripts. **[MEASURED]**: $\log S$ is linear in $\ln \ln x$ ($R^2 = 0.997$ for $x \geq 10^6$); the selection constant climbs $\widehat{C}(80) : 1.57 \rightarrow 2.23 \rightarrow 2.69$ across $x = 10^6, 10^8, 10^9$.

c4_growing_A.py Growing- A scan at fixed $x = 10^9$, $A \leq 200$. **[MEASURED]**: the two central die-condition quantities of Section 4.8 — $\log \widehat{C}(A) \sim 0.104 \log^2 A$ on $A \geq 40$ ($R^2 = 0.998$), and $\text{cond}(a) \rightarrow 1$ past $a = 80$ (mean 0.993 on $a = 161$ –200).

c4_growing_box_probe.py An earlier, coarser probe of the same quasi-independence question in both the $d = 1$ -growing- A and 2-dimensional (a, d) -growing-box directions, predating and superseded in detail by the two scripts above.

c4_two_shift_probe.py Pairwise independence: $\rho_{ab}/(\rho_a \rho_b)$ for $1 \leq a < b \leq A \leq 120$. **[MEASURED]**: consistently 1.000 ± 0.05 — the empirical basis for the programme’s repeated methodological point that pair-level statistics are the “wrong moment” for T(A), since the full A -fold intersection can drift while all pairs stay independent.

c4_euler_factor.py The prime-aligned local Euler factor $C_{\text{euler}}(A)$ of Section 4.8, computed over primes $q \leq 10^7$. **[MEASURED]**: flat at ≈ 0.91 (the $1/q$ -normalised column) from $A = 40$ to $A = 200$, with an exhaustive residue-collision check finding *zero* collisions for $A \leq 200$.

c4_composite_layer.py The genuine-extra composite Euler product $\Pi_{\text{gen}}(a, \sqrt{x})$ (composite aligned divisors with no aligned prime factor), isolating the “2.7–3.0 $\times$ totient mismatch” between the prime-aligned and full **ClassRough** marginals. **[MEASURED]**: matches the empirical ratio $\rho_a^{\text{CR}}/\rho_a^{\text{prime}}$ to within 3% at $A = 40$ across $x = 10^6, 10^8, 10^9$.

c4_certificates.py The certificate-converse and shared-certificate “dummy covering” tests of Section 4.2. **[MEASURED]**: converse holds for only 84.6% of survivors at $x = 10^6$, $A \leq 40$ (exact at $\omega = 1$, falling to 35% at $\omega = 4$); certificates inherited from divisors of a cover only $\approx 73\%$ of later survivors, ruling out shared-certificate inheritance as the explanation for $\text{cond}(a) \rightarrow 1$.

`c4_defect_layer.py` Exact enumeration (or Monte Carlo beyond a size cutoff) of the combinatorial defect between ClassRough and the certificate union, confirming that the two coincide exactly on $a \in \{1, 2, 3, 6\}$ (the elementary 2-group locus $\lambda(4a) \mid 2$) and diverge starting at $a = 4$, matching Section 4.2.

`c4_c2_symbols.py` The C2 cross-moduli Jacobi-symbol correlation test of Section 4.9, at $x = 10^9$, $A \leq 80$, 564 survivors. [MEASURED]: correlations consistent with independence; wrong sign to explain $\widehat{C} > 1$.

`c4_surface_fit.py` Fits the two-dimensional surface $\widehat{C}(A, x)$ jointly in A and x from the JSON outputs of the scripts above, isolating the x -slope deficit ($\kappa_{\text{eff}} \approx 0.121$ at $A = 80$) from the A -curvature ($\kappa_{\text{eff}} \approx 0.035$ at $x = 10^9$) as genuinely different partial derivatives of $\log S$, and tracking $c'(x)$ across $x = 10^6, 10^8, 10^9$ ($0.063 \rightarrow 0.076 \rightarrow 0.094$, still rising, with a naive linear extrapolation to κ landing around $x \sim 10^{24}$ — explicitly flagged as *not* a reason to run further large-scale computation).

`c4_sieve_constant.py` Computes the Selberg constant $C_{\text{sieve}}(A)$ and its Mertens-constant component $B(A)$ of $T(A)^+$ (Section 4.6), cross-checked at cutoffs $Q = 10^6$ and $Q = 10^7$.

`c4_t3_euler.py` The four certificate-assignment Euler products specific to $T(3)$ (Section 4.3), each computed to $q \leq 10^6$, together with the empirical occupancy of the four assignments among 2,370 hard primes at $x = 10^6$ (exact match to the kernel theorem’s prediction: 656 class-rough on all three slices, 656 in the certificate union).

9.2 The E_{power} second-moment family

`e_power_suen_moments.py` Checks Lemma SM’s implied ratios R_{Δ}, R_{δ} on the Lean covering cells at $A \in \{24, 48, 80, 120, 160, 200, 240\}$ against several hub cutoffs T ; establishes the schedule $T \approx 5\mu$ used in the final theorem.

`e_power_suen_moments_large.py` A two-pass, memory-efficient re-implementation extending the same check to $A \in \{400, 800, 1200, 1600, 2000\}$, confirming that R_{Δ} settles near 0.13 and the Suen loss stays under 0.15μ throughout a further order of magnitude in A (Section 4.7).

9.3 The $T(3)$ -completion range-check family

Four scripts, all deliberately *range checks* rather than covering runs, support the four negative results of Section 4.4:

`t2_zheng_ranges.py` Evaluates Zheng’s piecewise exponent functions $\mathcal{L}(\theta)$ on a 181×181 grid of the switched divisor-exponent plane, computing the fraction of the plane covered by ordinary Bombieri–Vinogradov, by Zheng’s Theorem 1.1, and by neither; also implements the “uneven leftover” reroute bookkeeping of Section 4.4(i).

`t3_delta_ranges.py` Implements the Farey- k incompatibility lemma of Section 4.4(ii) and checks it symbolically across the three named cells (BV wall, uneven, symmetric).

`t3_kloosterman_ranges.py` Computes the admissible-saving windows of Pascadi, Blomer–Pascadi and MQW as functions of the stall parameter δ , locating the first genuine saving points cited in Section 4.4(iv).

`t3_monodromy_ranges.py` Checks the FKMS length hypotheses (Theorems 1.3/1.4) against the three named cells, confirming the mismatch tabulated in Section 4.4(iii): the one cell where the inequalities hold cannot carry a prime modulus after switching, and the one cell that can has Type II lengths too long for either theorem.

9.4 Geometric and combinatorial searches

`s1_schinzels_separators.py` The Schinzel-separator sweep of Section 5.7: exhaustive complete search for positive solutions ($x \leq 3n/m$) and a bounded search (coordinates ≤ 60) for mixed-sign witnesses, over all reduced pairs $1 < m/n < 3$, $n \leq 40$; reproduces the 86-separator count.

`c7_bounded_coupling_search.py` Two bounded computations for candidate C7 (Section 7): an exhaustive search over degree-1 Witness identities with modulus ≤ 30 and coefficients ≤ 4 (193 identities found, none covering a hard class), and a coupling-mass comparison between the full Type-I covering family and its bounded- $\omega(q)$ sub-slice at $A \leq 64$, finding no evidence of the mass concentration C7 would need.

9.5 Methodological remarks

Three features of the computational programme deserve note as part of its overall epistemic discipline. First, every script’s docstring states explicitly what the computation does *not* establish — e.g. `c4_S_xscan.py`’s header notes it neither proves nor refutes H_ES, and `s1_schinzels_separators.py`’s header notes it does not check Brauer–Manin pairings. Second, several scripts are explicitly framed as adjudicating between two or more competing predictors for the same quantity (e.g. `c4_surface_fit.py` comparing $\log^2 A$, $\sum \log n_\chi/a$, $\log Z_{\text{global}}$, and $\sum \log n_\chi$ as candidate formulae for $\log \hat{C}(A)$, finding $\log^2 A$ the best fit by R^2 while explicitly declining to treat this as more than curve-fitting). Third, the repeated instruction “do not run $x = 10^{10}$ ” recurs verbatim across a dozen documents and script docstrings, functioning as an explicit acknowledgement that the open die-conditions of Sections 4.2 and 4.8 cannot be resolved by further computation at reachable scale, and that the correct next step is theoretical (Euler-product-level) rather than empirical.

10 Discussion and assessment

10.1 What has been achieved

Judged against its own stated objectives (Section 1), the programme delivers five concrete, independently assessable items. First, a genuine machine-checked reduction: the Erdős–Straus conjecture is formally reduced, inside the Lean kernel and with an explicit, published axiom audit, to a single named analytic proposition (`AnalyticSurvivorBound`) together with a finite verified range now standing at $n < 100,000$. This is not a new mathematical fact — the reduction to the six hard classes mod 840 is classical — but its complete, axiom-tracked formalisation, including the covering system and the divisor-form duality, is a genuine piece of infrastructure not previously available. Second, a real unconditional theorem: Theorem 4.9 improves the 1970 Vaughan exceptional-set bound from an $\exp(-c(\log x)^{2/3})$ saving to a genuine power saving $O(x^{1-\delta})$, via an elementary, fully written (if not machine-checked) second-moment argument. Third, a substantial and unusually disciplined formalisation of the Bright–Loughran Brauer–Manin theory, including a from-scratch finite verification of the arithmetic heart of their Lemma 3.8 and an extensive discharge of the Hilbert reciprocity the ES problem actually needs. Fourth, and arguably the programme’s most distinctive contribution, a coherent explanation — the k-budget invariant — of *why* the classical approaches have stalled, instantiated concretely five separate times (the fundamental lemma, the large sieve, Vaughan’s exponential sums, a circuit-complexity fooling argument, and the four-to-five T(3)-completion look

investigations), each converging on the same obstruction from a different technical direction, with Section 4.5’s capstone document giving the clearest single statement of why. Fifth, a disciplined external triage (Section 6): a sibling covering-density effort is checked against the programme’s own k -budget diagnosis and found to be another instance of it rather than an escape, and an external claimed proof is checked directly against the classical Mordell obstruction and shown not to establish the covering statement it needs — in both cases with the specific point of failure named rather than the claim being dismissed by assertion.

10.2 What has not been achieved, stated plainly

The programme is explicit, at every level of its own documentation, that it has not proved the conjecture, has not reduced it to a statement the mathematical community regards as tractable, and has not discharged either of its two named open interfaces. Three specific limitations are worth isolating for a reader assessing the work’s significance. First, the k -budget invariant itself is explicitly labelled “conjectural, proof-shaped” rather than a theorem; it is a strong organising heuristic, well corroborated by five independent instantiations, but it is not established that *every* conceivable interval-intrinsic method is capped by it. Second, the $T(A)/T(3)$ selection-constant question — whether $C(A)$ grows sub-critically against $\exp(\kappa \log^2 A)$ — remains open with a thin empirical margin ($c' = 0.104$ against $\kappa = 0.139$ at the largest tested scale) and an explicitly acknowledged live risk (“dummy covering”, $\text{cond}(a) \rightarrow 1$) that the programme itself regards as the more pressing concern. Third, the geometric route’s terminal finding — that every mechanism tried reduces to the analytic core — while a genuine and carefully earned negative result, means Route 2 does not currently supply an independent path to the conjecture; its value is prospective (an effectivity framework for strongly-obstructed log K3 surfaces) rather than realised.

10.3 On the methodology

Two methodological features of the archive merit comment independent of the mathematical content. First, the discipline of recording refuted intermediate claims rather than silently correcting them (most visibly in the §4f/§4h correction of the original, erroneous Vaughan-exponent “barrier”, and in the $C5/C5_1/C5_2$ refutation chain) makes the research log unusually auditable: a reader can verify not only what is currently claimed but what specific error the current claim corrects, and why. Second, the explicit “Gate A” discipline (Section 4.8) separating a written, useful record (E_power) from a machine-checked claim against the load-bearing interface is a genuine safeguard against the most common failure mode of ambitious computational mathematics programmes — the silent inflation of a genuine partial result into an implicit claim of having solved the real problem. Both practices are followed consistently throughout the archive and are reflected, where relevant, in the status tags used throughout this report.

10.4 Summary

In summary: the archive contains no proof of the Erdős–Straus conjecture and does not claim one. It contains a substantial, carefully audited Lean 4 formalisation of the classical reduction and of a non-trivial slice of the Bright–Loughran geometric theory; a new unconditional exceptional-set bound stronger than the 55-year-old published record; a precisely argued (if not fully proved) explanation of the structural obstruction the conjecture presents to every method tried; and four honestly negative results checking specific, currently published 2025–26 analytic number theory against that obstruction. Considered as a piece of research infrastructure and a research log, it is unusually complete and unusually disciplined about the distinction between what is proved, what is claimed, what is measured, and what has been tried and found not to work.

A Complete archive file catalogue

This appendix lists every file in `erdos-strauss-main` (87 files), grouped by type, with a one- or two-line description and a cross-reference to the section of this report discussing it. File sizes are as extracted from the archive.

A.1 Lean source files (13)

File	Lines	Contents	§
<code>ErdosStraus.lean</code>	3420	Core reduction, covering system, analytic interfaces, <code>ClassRough</code> /certificate theory, elementary $d = 1$ Henriot-interface sieve.	3.4
<code>BrightLoughran.lean</code>	407	Serre 2-adic symbol data, finite heart of BL Lemma 3.8, <code>InvariantData</code> interface, Theorem 1.2 anatomy, $p = 1009$ instances.	3.5
<code>ErdosStrausQR.lean</code>	170	Mathlib discharge: <code>IsPrime</code> ↔ <code>Nat.Prime</code> , Yamamoto fingerprint, quadratic reciprocity.	3.6
<code>ErdosStrausBLRoute.lean</code>	3675	$\mathbb{Z}$ -model, box bound, octant invariant, <code>TubEpHypothesis</code> , extensive Hilbert reciprocity discharge.	3.7
<code>ConicFiber.lean</code>	194	Split-fibre identity and exact divisor correspondence, all $n \geq 2$.	3.8
<code>TwistDescent.lean</code>	219	Real-compatible twist $V_{n,d}$; Gate 2/Gate 3; the $C5_2$ refutation.	3.9
<code>SchinzelSep.lean</code>	84	Signed 9/5 point; ordered no-positive-solution proof.	3.10
<code>SchinzelDecide.lean</code>	268	Decidable positive-octant solver with soundness/completeness; 9/5, 13/7, 4/5 instances.	3.10
<code>NoVieta.lean</code>	156	Linearity of the affine equation; uniqueness; no Vieta involution.	3.11
<code>Leochlon.lean</code>	95	Type-II identity dictionary $(4b - 1)(4c - 1) = 4p\delta + 1 \rightarrow \text{ES.IsES}$.	3.12
<code>Stormer.lean</code>	855	Størmer’s theorem on consecutive smooth numbers (ported, Apache 2.0, library only).	3.13
<code>archive/ErdosStraus-core.lean</code>	2073	Frozen earlier snapshot of the covering file; not a Lake root.	3.14
<code>archive/ErdosStraus-mathlib-project.lean</code>	2480	Duplicate of <code>ErdosStraus.lean</code> .	3.14

A short `archive/README.md` (not counted as Lean; see Appendix [A.3](#)) accompanies the two frozen files and states explicitly that they are not Lake roots.

A.2 Build configuration and licensing (9)

- `lakefile.toml` — package `ErdosStraus`; declares the Mathlib dependency (pinned commit `1a0ef26d...`) and the eleven-file Layer-A/B library root list.
- `lean-toolchain` — pins `leanprover/lean4:v4.34.0-rc1` (documented as a release candidate, not a moving nightly).
- `lake-manifest.json` — resolved dependency manifest generated by `lake`; `CONTRIBUTING.md` directs that it be committed with any toolchain or Mathlib bump.
- `.gitignore` — excludes `.lake/`, `.oleans/`, `*.olean`, `*.zip`, `__pycache__`.

- `.github/workflows/ci.yml` — GitHub Actions workflow; runs `lake exe cache get` and `lake build` against the pinned Mathlib cache on every push and pull request.
- `requirements.txt` — a single pinned `numpy` version constraint for the numeric scripts of Section 9.
- `CONTRIBUTING.md` — pinning policy table, build instructions, and an explicit restatement of scope (no fake theorem, no `sorry` on the QED line, library files are not progress toward the load-bearing interface).
- `LICENSE` — MIT licence, copyright Riley Betts Ltd, with an explicit carve-out noting that `Stormer.lean` remains Apache 2.0.
- `LICENSE-APACHE` — the Apache License 2.0 text covering `Stormer.lean` alone, copyright Alexander Benjamin Worth Burns and Ravi Bajaj.

A.3 Markdown documents (22)

File	Contents	§
<code>README.md</code>	Repository front matter: layer map, build instructions, axiom audit table, the “novel results (unverified)” table, the authorship/role statement, “what’s in Lean” / “what this does not claim” summaries, full bibliography.	3
<code>archive/README.md</code>	Short note: the two frozen Lean files are not Lake roots and should not be imported.	3.14
<code>erdos-strauss-programme.md</code>	The programme brief and repository front door: five open geometric questions, a chronological chain of dated postscripts (v0.9–v0.23) recording each investigation’s outcome, the authorship statement.	1.2,1.3
<code>erdos-strauss-novel-structures-plan.md</code>	The primary research log: eleven initial options considered, the programme (Tracks A/B/C), and sections §4a–§4w documenting the k-budget investigation, the Bright–Loughran route, and their synthesis.	4.1,5
<code>erdos-strauss-road-to-lean-qed.md</code>	The execution roadmap: roadmap Layer A/B/C, the cardinality-to-QED bridge, the Renewal Lemma, and the target Lean architecture.	3.1
<code>erdos-strauss-conjecture-spec.md</code>	The acceptance specification, requirements R1–R8, and the scorecard of Table 3.	8
<code>erdos-strauss-candidate-conjectures.md</code>	The full candidate list C1–C11, NV, S1, S1 ₁ , with status and primary risk for each.	7
<code>erdos-strauss-T-A.md</code>	T(A) working note: certificate architecture, defect layer, the two x-scan/growing-A measurements, the local Euler factor, T(A) ⁺ , E _{lane} .	4.2,4.6
<code>erdos-strauss-T-3.md</code>	T(3) statement, proof of T(3) ⁺ , the three-step lower-bound plan, and the r_χ -to-Kloosterman look (negative result (iv)).	4.3,4.4
<code>erdos-strauss-T-2.md</code>	The Zheng simultaneous-progressions look (negative result (i)).	4.4
<code>erdos-strauss-T-3-delta.md</code>	The Heath-Brown δ -method look (negative result (ii)).	4.4
<code>erdos-strauss-T-3-monodromy.md</code>	The FKMS joint-monodromy look (negative result (iii)).	4.4
<code>erdos-strauss-varying-modulus-gap.md</code>	Capstone document: the four (with an informal fifth) T(3)-completion looks classified into one common failure mode.	4.5

File	Contents	§
zheng-wellfactorable.md	Technical note on well-factorability of real characters (Zheng Def. 1.1), feeding <code>erdos-strauss-T-2.md</code> .	4.4
erdos-strauss-E-power.md	Statement and full written proof of Theorem 4.9 (E_power) and Lemma SM.	4.7
erdos-strauss-E-partial.md	Gate A write-up: what may/may not be claimed toward <code>AnalyticSurvivorBound</code> ; the schedule-retuning argument of Section 4.8.	4.8
erdos-strauss-gs-reformulation.md	The Granville–Soundararajan restatement of the survivor-density target; the C2 null result.	4.9
erdos-strauss-sieve-desk.md	The analytic question list.	4.10
erdos-strauss-loughran-orbit.md	The geometric question list: five numbered open questions in the Bright–Loughran literature.	5.10
erdos-strauss-prior-archive.md	Triage of the sibling Track-1 covering-density archive: what was kept as library material and what was left out.	6.1
erdos-strauss-bradford-lemma3.md	Examination of the Bello-Hernández-Benito-Fernández rigidity lemma and of Bradford 2026 as a claimed proof.	6.2
erdos-strauss-leochlon.md	Documentation of the Leochlon Type-II identity dictionary.	3.12

A.4 Python scripts and their JSON outputs (20 scripts, 23 JSON files)

Script	Purpose	Output JSON(s)	§
<code>c4_S_xscan.py</code>	x -scan of $S(A, x)$ at fixed A ; hard-prime sieve infrastructure reused throughout.	<code>c4_S_xscan_X1000000000.json</code> , <code>_X1000000000.json</code>	9
<code>c4_growing_A.py</code>	Growing- A scan at fixed $x = 10^9$; $\text{cond}(a)$ and $\widehat{C}(A)$.	<code>c4_growing_A_X1000000000_A200.json</code>	9
<code>c4_growing_box_probe.py</code>	Earlier two-directional independence probe.	<code>c4_growing_box_X1000000.json</code> , <code>_X1000000.json</code>	9
<code>c4_two_shift_probe.py</code>	Pairwise-independence and C2 Jacobi-symbol probe.	<code>c4_two_shift_probe.json</code>	9
<code>c4_euler_factor.py</code>	Prime-aligned local Euler factor $C_{\text{euler}}(A)$ and residue-collision check.	<code>c4_euler_factor_Q1000000_A200.json</code>	9
<code>c4_composite_layer.py</code>	Genuine-extra composite Euler product Π_{gen} .	<code>c4_composite_layer_Z50000_A80.json</code>	9
<code>c4_certificates.py</code>	Certificate converse and shared-certificate dummy-covering tests.	<code>c4_certificates_X1000000_A40.json</code>	9
<code>c4_defect_layer.py</code>	Exact/Monte-Carlo defect between <code>ClassRough</code> and the certificate union.	<code>c4_defect_layer_A12_W5.json</code>	9
<code>c4_c2_symbols.py</code>	C2 cross-moduli Jacobi-symbol correlation test.	<code>c4_c2_symbols_X1000000_A30.json</code> , <code>_X1000000000_A80.json</code> , <code>_X1000000000_A80.json</code>	9
<code>c4_surface_fit.py</code>	Joint (A, x) surface fit of $\widehat{C}$; predictor comparison.	<code>c4_surface_fit.json</code>	9

Script	Purpose	Output JSON(s)	§
c4_sieve_constant.py	Selberg constant $C_{\text{sieve}}(A)$ and $B(A)$ of $T(A)^+$.	c4_sieve_constant_Q1000000_A200.json, Q1000000_A200.json	9
c4_t3_euler.py	$T(3)$ -specific certificate-assignment Euler products and occupancy.	c4_t3_euler_Q1000000.json	9
e_power_suen_moments.py	Lemma SM ratio checks, $A \leq 240$.	e_power_suen_moments_A240.json	9.2
e_power_suen_moments_large.py	Lemma SM ratio checks extended to $A \leq 2000$.	e_power_suen_moments_A2000.json	9.2
t2_zheng_ranges.py	Zheng $\mathcal{L}(\theta)$ range check (negative result (i)).	t2_zheng_ranges.json	4.4
t3_delta_ranges.py	Farey- k incompatibility check (negative result (ii)).	t3_delta_ranges.json	4.4
t3_kloosterman_ranges.py	Pascadi/Blomer-Pascadi/MQW saving-window check (negative result (iv)).	t3_kloosterman_ranges.json	4.4
t3_monodromy_ranges.py	FKMS gallant-monodromy length check (negative result (iii)).	t3_monodromy_ranges.json	4.4
s1_schinzels_separators.py	Schinzels narrow-band separator sweep.	(no separate JSON; prints results)	5.7
c7_bounded_coupling_search.py	C7 identity search and coupling-mass comparison.	(no separate JSON; prints results)	9.4

The count of 23 JSON files above excludes `lake-manifest.json` (Appendix A.2), which is a build artefact rather than an experimental output.

B The axiom audit

Table 7 reproduces, for reference, the axiom audit table maintained in `README.md`, obtained by `#print axioms` on Lean 4.33.0 for every named theorem on the file-map Layer-A kernel-checked spine. `native_decide` instances add a generated reduction axiom (`Lean.ofReduceBool`, materialising as `_.native.native_decide.ax_1_1` on 4.33.0) and are listed separately.

The five `native_decide` theorems, each additionally carrying `Lean.ofReduceBool`, are: `primes_below_10000_covered`, `hard_primes_below_10000_covered` and `okay_below_100000` in `ErdosStraus.lean`, and `thm12_instance_1009`, `witness_fingerprint_1009` in `BrightLoughran.lean`.

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Theorem	Axioms
witness_sound	propext, Quot.sound
es_1009	propext, Quot.sound
bl_thm12_prime_anatomy	propext, Quot.sound
primes_below_100_covered_kernel	none (decide)
lemma38_check_true	none (decide)
conditional_qed_hard	propext, Quot.sound, Classical.choice
lemma32	propext, Quot.sound, Classical.choice
ConicFiber.fiber_identity	propext, Quot.sound
ConicFiber.fiber_to_divisor	propext, Quot.sound
ConicFiber.divisor_to_fiber	propext, Quot.sound
TwistDescent.descent	none
TwistDescent.naive_cover_empty	propext, Quot.sound
TwistDescent.lift	propext, Quot.sound, Classical.choice
TwistDescent.twist_real_iff	propext
TwistDescent.t_fiber_section_cleared	propext
TwistDescent.u1_fiber_section_cleared	propext
TwistDescent.u1_fiber_twist	propext
SchinzelSep.signed_9_5	none
SchinzelSep.no_pos_9_5	propext, Quot.sound
SchinzelDecide.sep_9_5	none (decide)
SchinzelDecide.sep_13_7	none (decide)
SchinzelDecide.pos_4_5	none (decide)
SchinzelDecide.decideCplus_sound	propext, Quot.sound
SchinzelDecide.decideCplus_complete	propext, Quot.sound
SchinzelDecide.no_pos_9_5	propext, Classical.choice, Quot.sound
SchinzelDecide.no_pos_13_7	propext, Classical.choice, Quot.sound
NoVieta.unique_x	propext, Quot.sound
NoVieta.es_unique_x	propext, Classical.choice, Quot.sound

Table 7: Axiom audit of the kernel-checked spine (README.md).

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Bright–Loughran geometry (Route 2).

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